Zooplankton diversity partitioning: implications for stream conservation in the Atlantic Forest

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Abstract

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1 Introduction

The Atlantic Forest, recognized for its species richness and high level of endemism, is one of the most threatened tropical forests in the world, making it a global priority for biodiversity conservation (Fonseca 1985, Myers et al. 2000, Tabarelli et al. 2010). Originally, its vegetation cover extended over 1.5 million km^2, however, centuries of human occupation and resource exploitation have led to severe landscape reduction, compromising the ecological integrity of its remnants and the maintenance of biodiversity (Tabarelli et al. 2010, Safar et al. 2020, Vancine et al. 2024). Habitat loss driven by deforestation is directly linked to forest fragmentation,

which promotes the isolation of remnants, reduces landscape connectivity, and restricts species dispersal (Zaú 1998, Tabarelli et al. 2010, Souza et al. 2025).

These impacts are not confined to terrestrial environments, they extend to the associated aquatic ecosystems and, most importantly, headwater streams (Malabarba 2006, Meyer et al. 2007). Forest fragmentation and land management can generate significant impacts on stream fauna within the Atlantic Forest, as aquatic organisms are highly sensitive to changes in natural water conditions (England and Rosemond 2004). Furthermore, landscape-level processes—such as disturbance, fragmentation, dispersal, and mosaic structure—exert a direct influence on local populations (Maltchik and Medeiros 2006, Lôbo et al. 2011). Therefore, understanding the ecological processes structuring freshwater communities is essential to comprehend how diversity is distributed across different spatial scales (Ávila et al. 2011) and to evaluate the ecological health of stream.

In addition to landscape alterations, stream dynamics are strongly driven by seasonal factors. The Atlantic Forest features a predominantly tropical climate with rainfall distributed throughout the year. These hydrological fluctuations directly influence Neotropical freshwater environments by modifying physical and chemical conditions (Barboza et al. 2024). The shift between dry and rainy seasons alters resource availability, microhabitat structure, and the establishment and persistence of organisms, thereby shaping community composition and dynamics (Ward 1998, Souza et al. 2025). During the dry season, reduced precipitation and lower streamflow stabilize hydrological conditions, which may favor the persistence of local populations. Conversely, the wet season is characterized by increased rainfall and discharge, intensifying disturbance events, dispersal, and species turnover. Consequently, seasonality acts as a major driver of aquatic community organization, making these organisms valuable biological tools for assessing the ecological state of freshwater systems (Souza et al. 2025), and shifts in species composition and community turnover often reflect underlying alterations in environmental conditions (Keppeler and Hardy 2004, Melo and Medeiros 2013, Shell and Collier 2018, Freiry et al. 2020).

Within this context, the design of efficient protected areas that incorporate aquatic systems, ought to rely on a robust network of interconnected spatial scales and meaningful observation of species distributions within and surrounding the areas of biological interest (Maltchik and Medeiros 2006, Herbert et al. 2010, Socolar et al. 2016). Essential to this is the choice of relevant species, capable of representing the central elements of the biological components of lotic communities. Zooplankton, particularly Rotifera, Cladocera, and Copepoda, play a pivotal role in stream ecosystems (Santos et al. 2024, Amaral et al. 2024). These organisms exhibit diverse body sizes, morphologies, life histories, and functional roles, serving as excellent bioindicators due to their rapid sensitivity to environmental changes (Bonecker et al. 2013). Furthermore, zooplankton constitute a critical food source for fish and other organisms, an essential trophic bridge that transfers energy from primary producers to higher trophic levels (Balkić et al. 2022, Marcelino et al. 2025, Souza et al. 2025, Altindağ and Berdï 2025).

Furthermore, in recent decades, researchers have argued that biodiversity assessments should not be reduced to a single diversity index. Romanelli et al. (2022) proposes that a multiple

statistical diversity approach with multi-taxon evaluation provides a more comprehensive understanding of environmental ecological health and species responses to anthropogenic stress. Nevertheless, few approaches allow for the assessment of diversity organization across explicit hierarchical spatial scales, such as in streams. Diversity partitioning stands out in this context. This framework decomposes overall diversity across multiple spatial levels, testing whether observed diversity is higher or lower than expected under a null distribution model (Lopes et al. 2014), thereby offering valuable insights into community assembly and underlying ecological drivers.

Among the diversity partitioning framework, additive partitioning is widely applied in ecological studies, even though, Ligeiro et al. (2010) point out that its application in freshwater systems remains relatively scarce and highlights a critical knowledge gap regarding alpha and beta diversity distribution in freshwater systems. Additive diversity partitioning has been shown to provide a robust framework to evaluate how freshwater communities respond to landscape fragmentation and/or seasonal dynamics (Akhyar et al. 2026). By decomposing total diversity into hierarchical α and β components, it is possible to quantify spatial contributions to overall diversity and detect species turnover across scales (Jost 2007). That is, under the additive partitioning model, total regional diversity (γ-diversity) is partitioned into additive alpha (α) and beta (β) components (Crist et al. 2003), where α-diversity represents the mean local richness and β-diversity reflects spatial turnover or variation in species composition among sampling units (Whittaker 1972, Jost 2007, Ligeiro et al. 2010, Ávila et al. 2011, Baselga 2012).

Therefore, understanding the spatial structure and distribution of biodiversity across scales remains a central theme in community ecology and conservation planning (Lande 1996, Crist et al. 2003). These approaches are based on the concepts of species turnover and nestedness, two distinct components used to break down and understand beta diversity – which measures how much species composition changes between different communities or sites (Baselga 2010, 2012). Total regional species richness (γ-diversity) can be hierarchically partitioned into its additive components: local richness (α-diversity) and species turnover among assemblages (β-diversity) across spatial or temporal scales (Lande 1996). Examining these components allows ecologists to disentangle whether regional diversity is primarily governed by fine-scale local processes or broader spatial heterogeneity (Crist et al. 2003, Tuomisto 2010). Beta diversity itself reflects the variation in species composition among sites and is driven by two distinct ecological phenomena: turnover (species replacement) and nestedness (richness differences) (Baselga 2010). Species turnover occurs when species are replaced by different species along environmental or spatial gradients, usually driven by environmental filtering, competitive exclusion, or dispersal limitation (Baselga 2010, Tuomisto 2010). In contrast, nestedness occurs when communities at species-poor sites are non-random subsets of those at species-rich sites (Baselga 2010). Nested patterns typically stem from deterministic processes such as selective extinction, differential colonization capacity, or habitat fragmentation that sequentially eliminate sensitive taxa while retaining tolerant generalists (Veech et al. 2003, Baselga 2010).

Evaluating the relative contribution of turnover and nestedness—alongside null models using

Standardized Effect Sizes (SES)—provides critical insights into assembly rules and conservation prioritization (Gotelli and McCabe 2002, Gering et al. 2003). A pattern dominated by turnover (SES > 0) implies strong ecological differentiation and high beta diversity, signaling that distinct localities host unique biological assemblages (Baselga 2010). From a conservation perspective, high turnover necessitates a network-wide preservation strategy, as multiple sites must be protected to encompass regional biodiversity (Baselga 2010). Conversely, a system dominated by nestedness or fine-scale homogenization (SES < 0) indicates that local assemblages are nested sub-collections of regional species pools. Under a nested regime, management efforts are most effectively concentrated on protecting the largest or highest-quality species-rich sites, which inherently harbor the vast majority of the regional pool (Baselga 2010).

This study describes the richness and density of zooplankton, comparatively between two small head water streams subject to different levels of environmental pressures. One originates withing a protected area - Biological Reserve, or REBIO, according to BRASIL (2000) - and its course extends beyond its administrative boundaries (MMA 2003), and a second one that flows entirely outside of the protected area. Within this framework, we use hierarchical additive partitioning to evaluate the relative contribution of different spatial and temporal scales to the diversity of zooplankton assemblages. Given the importance of the Atlantic Forest to the biodiversity (Pereira et al. 2026) and its fragmented stated north of the São Francisco river basin (Silva and Casteleti 2003), we aim with this study, (1) to infer environmental degradation from species turnover (Baselga 2010) and (2) to associate beta-diversity to the ecological health of sites inside and surrounding a protected area of Atlantic Forest. Given the expected different levels of environmental pressures in the area (Gouveia et al. 2017), we hypothesize that the diversity is not be homogeneously distributed among the different spatial scales but concentrated at the finer levels and seasonal variation will be limited to species composition, but no spacial variation is expected between dry and wet seasons. Some environmental features (reach morphology, habitat structure, and water quality) are also described to explore possible explaining factors for the observed patterns. Ultimately, this study intends to generate key baseline data on the ecological condition of the studied streams and the ecological mechanisms structuring zooplankton communities, contributing to optimal conservation strategies in Atlantic Forest streams.

2 Material and methods

2.1 Study area and sampling design

This study was conducted in the Guaribas Biological Reserve (REBIO Guaribas), a protected area of Atlantic Forest in northeastern Brazil. This protected area is located in a transitional zone between the Caatinga and Atlantic Forest ecoregions. The land use of the study area is based on the occupation of traditional human populations and an increasing use of land for large scale monocultures, mostly sugar cane (Heinrichs et al. 2017). Other than these, small scale agricultural activities, fishing, and the modification of the natural vegetation for

livestock are important anthropogenic pressures that affect the buffer zone of the REBIO (Arruda et al. 2013, MMA 2026). Other common uses of the soil in the area are, coconut plantations, orchards and subsistence agriculture. These land uses represent important sources of pollution, either through agrochemicals runoff or direct and indirect domestic effluents (such as septic tanks). These nonpoint source pollutants contaminate underground water and create a spatial gradient of environmental disturbance (Mehla et al. 2026). Therefore, the mainly rural nature of the land use associated with the management of vegetation, will cause direct impact to aquatic systems, by the siltation of streams and contamination of groundwater. The hydrographic network of the study region is composed predominantly of small catchments containing first- and second-order streams (Strahler 1957 classification) (Figure 1). These watercourses are tributaries of the Camaratuba River and are sustained by numerous springs distributed throughout the region. The two main stream systems in the study area are the Barro Branco and Caiana catchemnts. The Barro Branco stream is approximately 8.8 km long and originates within the protected area. Its upper section remains within the boundaries of the REBIO, whereas its middle and lower sections extend into the surrounding landscape. Despite this transition, riparian vegetation is preserved along most of the stream. The Caiana stream is approximately 9.8 km long and also originates within the REBIO, although most of its course lies outside the protection area.

The climate in the region is classified as As' (Peel et al. 2007), characterized by a mean annual temperature of approximately 25 °C, with maximum temperatures occurring mainly between February and March, during the dry period, as well as by low annual thermal amplitude. Annual precipitation ranges from 1,400 to 1,800 mm, concentrated between March and July, whereas the lowest rainfall levels occur between September and December. In the study area, rainfall is concentrated between March and July, beginning in autumn and extending into winter. This pattern is attributed to the influence of cold air masses from the South Atlantic ocean that affect the region. The hydrographic network is composed of small headwater streams fed by rain (Gouveia et al. 2017).

A total of 15 sampling sites distributed longitudinally across the two Atlantic Forest streams were selected to capture spatial variation in environmental conditions and habitat characteristics. The Barro Branco stream included sampling sites located within a protected area (REBIO Guaribas) and sites located outside its boundaries, whereas all sampling sites in the Caiana stream were located outside the protected area. This sampling design allowed the assessment of zooplankton diversity across nested spatial scales and contrasting conditions of environmental protection and human influence (Figure 1). Sampling included temporal (dry and wet seasons) and spatial scales. The spatial scales of zooplankton diversity were represented by the total gamma ( ) diversity, encompassing both study streams. At each stream, several sites were selected according to their position in the main channel of the catchment basin and, at each site, three habitas were sampled to represent different habitat types (pool, riffle, near underwater structures) (Frissell et al. 1986). We also considered a further spatial scale being the upper reaches (portion of the Barro Branco stream within the conservation unit and the Caiana stream before its confluence with the Pipina stream) and the lower reaches (portion of the Barro Branco stream outside the conservation unit and the Caiana stream after its confluence with

the Pipina stream) (Figure 1). Sampling was conducted during the dry (January-February) and wet (June-July) seasons of 2024.

2.2 Environmental data collection

Environmental conditions were characterized using four groups of variables, following procedures adapted by Medeiros et al. (2008) from the protocols proposed by Pusey et al. (2004) and Mugodo et al. (2006): physicochemical water quality, stream site morphology, substrate composition, and marginal habitat structure. Water quality was characterized using physical and chemical measurements obtained with portable instruments, including water temperature (°C) and dissolved oxygen concentration (mg/L). Turbidity was subsequently measured in the laboratory with a turbidimeter using water samples collected at each sampling site. Current velocity (m/s) was estimated using the float method, as described by Maitland (1990). Morphological characteristics were assessed from three randomly selected transects at each site, where stream width (cm) and water depth (cm) were measured. Bank slope was classified into four categories according to its inclination: 1, 20°; 2, > 20–30°; 3, > 30–50°; and 4, > 50°. Broader-scale landscape characteristics, including elevation and stream length, were obtained from handheld GPS records and satellite imagery. Substrate composition and the physical structure of marginal habitats were evaluated from 9 to 12 quadrats (1 m^2) distributed along the terrestrial-aquatic interface at each sampling site. Within each quadrat, the proportional cover of different substrate categories was visually estimated, including mud, sand, cobbles, fine and coarse gravel and rocks. The same quadrats were used to characterize littoral and submerged habitat features. Recorded habitat components included macrophytes, marginal grasses, leaf litter, attached and filamentous algae, overhanging and submerged vegetation, small and large woody debris, and root masses (Medeiros et al. 2008). For subsequent analyses and presentation, environmental measurements were summarized as mean values for each sampling site.

Three 250 mL replicate samples of water from each site were taken and refrigerated for concentration analysis (g L^{1}) of ammonia, nitrite, nitrate, orthophosphate (soluble reactive phosphorus) and total phosphorus. Nutrient concentrations were obtained based on Standard Methods for the Examination of Water (APHA 1998) and Mackereth et al. (1978).

2.3 Zooplankton sampling

Zooplankton was sampled at habitats within each sampling site, hereafter referred to as habitat samples. Collections were performed in the main channel of the streams at the surface water. At each sampling habitat, a plankton net (30 cm mouth diameter, 70 cm length, and 60 m mesh size) was used to filter 50 L of water. The net was rinsed between successive samples to prevent clogging and carryover of organisms. Each sample was concentrated to a final volume of 80 mL for laboratory analyses. Following collection, zooplankton samples were anesthetized with commercial sparkling water and subsequently preserved in 4% formalin (EPA 2020). Sucrose

was added to the preserved sample to prevent female cladocerans from losing eggs and to minimize carapace distortion (Haney and Hall 1973). In the laboratory, samples had their zooplankton identified and counted using a Sedgewick–Rafter counting chamber. The analyses focused on representatives of Rotifera, Cladocera, and Copepoda. Taxonomic identification was based on specialized literature and confirmed by specialists. To evaluate whether the sampling effort adequately represented the zooplankton assemblage, species accumulation curves were calculated (Melo 2014, Lucena et al. 2015).

2.4 Statistical analyses

Environmental variables were screened for multicollinearity and to identify redundant descriptors. Variables representing similar environmental information were also combined when appropriate, thereby reducing redundancy in the dataset (Legendre and Legendre 1998).

```
#ZOOREBIO23_PART - ORGANIZANDO DADOS AMBIENTAIS----
#####....----

dev.off() #apaga os graficos, se houver algum
rm(list=ls(all=TRUE)) #limpa a memória
cat("\014") #limpa o console

##CARREGANDO MATRIZES BRUTAS----

library(openxlsx)
habitat <- read.xlsx("dados/rebio23-habitat.xlsx",
                     rowNames = T,
                     colNames = T,
                     sheet = "ambiente",
                     startRow = 2)

# List columns with at least one NA
colnames(habitat)[colSums(is.na(habitat)) > 0]
# Identify columns containing only NA values
all.na <- names(habitat)[colSums(!is.na(habitat)) == 0]
all.na <- c(all.na, "w.Temp_C", "w.DO_mg.L")
#all.na <- c(all.na, "n.Ammonia_NH3_ug.L", "n.Nitrite_NO2_ug.L",
#            "n.Nitrate_NO3_ug.L", "n.Orthophosphate_SRP_ug.L",
#            "n.Phosphorus_TP_ug.L")
# Remove those columns by name
m_hab_part <- habitat[, !names(habitat) %in% all.na, drop = FALSE]
# Removendo linhas por nome
rownames(m_hab_part)
```

```r
del_rows <- c("C-C-P11-H1", "C-C-P11-H2", "C-C-P11-H3",
              "S-C-P11-H1", "S-C-P11-H2", "S-C-P11-H3")
del_rows
m_hab_part <- m_hab_part[!(row.names(m_hab_part) %in% c(del_rows)),]
m_hab_part
# List columns with at least one NA
colnames(m_hab_part)[colSums(is.na(m_hab_part)) > 0]
#rownames(m_hab_part)[is.na(m_hab_part$m.Width_m)]
# Remove columns with at least one NA
m_hab_part <- m_hab_part[, colSums(is.na(m_hab_part)) == 0, drop = FALSE]

###REMOVENDO COLUNAS ZERADAS DE HABITAT----

sum <- colSums(m_hab_part)
sum
zero_sum <- names(which(colSums(m_hab_part) == 0))
zero_sum #nomes das colunas zeradas
m_part_cols <- m_hab_part[(colSums(m_hab_part) != 0)] #em != a exclamação inverte o sentido
zero_sum2 <- names(which(colSums(m_part_cols) == 0))
zero_sum2 #nomes das colunas zeradas
sum<-colSums(m_part_cols)
sum

t_grps <- read.xlsx("dados/rebio23-habitat.xlsx",
                    rowNames = T,
                    colNames = T,
                    sheet = "grupos",
                    startRow = 2)

###REMOVENDO LINHAS ZERADAS DE GRUPOS----

rownames(t_grps)
del_rows
t_grps <- t_grps[!(row.names(t_grps) %in% c(del_rows)),]
t_grps

###SEPARANDO NUTRIENTES----

nutri <- grep("^n\\.", colnames(habitat), value = TRUE)
nutri
nutrientes <- habitat[, nutri, drop = FALSE]
nutrientes <- nutrientes[, colSums(!is.na(nutrientes)) > 0, drop = FALSE]
nutrientes <- na.omit(nutrientes)
```

```r

##SALVANDO MATRIZES FINAIS----

write.table(m_part_cols, "res_ms/m_hab_part.csv",
            sep = ";", dec = ".", #"\t",
            row.names = TRUE,
            quote = TRUE,
            append = FALSE)
write.table(nutrientes, "res_ms/t_nutri.csv",
            sep = ";", dec = ".", #"\t",
            row.names = TRUE,
            quote = TRUE,
            append = FALSE)
write.table(t_grps, "res_ms/t_grps.csv",
            sep = ";", dec = ".", #"\t",
            row.names = TRUE,
            quote = TRUE,
            append = FALSE)
t_grps <- read.csv("res_ms/t_grps.csv",
                   sep = ";", dec = ".",
                   row.names = 1,
                   header = TRUE,
                   na.strings = NA)
t_nutri <- read.csv("res_ms/t_nutri.csv",
                  sep = ";", dec = ".",
                  row.names = 1,
                  header = TRUE,
                  na.strings = NA)
m_hab <- read.csv("res_ms/m_hab_part.csv",
                  sep = ";", dec = ".",
                  row.names = 1,
                  header = TRUE,
                  na.strings = NA)

#ZOOREBIO23_PART - ORGANIZANDO DADOS DA COMUNIDADE----
#####....----

dev.off() #apaga os graficos, se houver algum
rm(list=ls(all=TRUE)) #limpa a memória
cat("\014") #limpa o console

library(openxlsx)

```

```r
##CARREGANDO MATRIZES BRUTAS----

densidade <- read.xlsx("dados/rebio23-zoopl.xlsx",
                    rowNames = T,
                    colNames = T,
                    sheet = "densidade_hab_indL")
densidade[1:5,1:5] #[1:5,1:5] mostra apenas as linhas e colunas de 1 a 5.

##COLUNAS ZERADAS DE DENSIDADE----
sum <- colSums(densidade)
sum
zero_sum <- names(which(colSums(densidade) == 0))
zero_sum #nomes das espécies zeradas
m_part_cols <- densidade[(colSums(densidade) != 0)] #em != a exclamação inverte o sentido
zero_sum2 <- names(which(colSums(m_part_cols) == 0))
zero_sum2 #nomes das espécies zeradas
sum<-colSums(m_part_cols)
sum

###LINHAS ZERADAS DE DENSIDADE----
m_part2 <- as.data.frame(t(m_part_cols))
sum <- colSums(m_part2)
sum
zero_sum <- names(which(colSums(m_part2) == 0))
zero_sum #nomes das espécies zeradas
m_part_rows <- m_part2[(colSums(m_part2) != 0)] #em != a exclamação inverte o sentido
zero_sum2 <- names(which(colSums(m_part_rows) == 0))
zero_sum2 #nomes das espécies zeradas
sum<-colSums(m_part_rows)
sum
m_dens <- as.data.frame((t(m_part_rows)))

##SALVANDO MATRIZES FINAIS REMOVIDAS AS LINHAS/COLUNAS ZERADAS----

write.table(m_dens, "res_ms/m_dens.csv",
            sep = ";", dec = ".", #"\t",
            row.names = TRUE,
            quote = TRUE,
            append = FALSE)
m_dens <- read.csv(res_ms/"m_dens.csv",
                    sep = ";", dec = ".",
                    row.names = 1,
                    header = TRUE,
```

```
                    na.strings = NA)

#ZOOREBIO23_PART - MATRIZ DE CONTAGEM----
#####....----

library(openxlsx)
contagem <- read.xlsx("dados/rebio23-zoopl.xlsx",
                      rowNames = T,
                      colNames = T,
                      sheet = "rebio23_80ml",
                      startRow = 4)
contagem[1:5,1:5] #[1:5,1:5] mostra apenas as linhas e colunas de 1 a 5.

###COLUNAS ZERADAS DE CONTAGEM----
sum <- colSums(contagem)
sum
zero_sum <- names(which(colSums(contagem) == 0))
zero_sum #nomes das espécies zeradas
m_part_cols <- contagem[(colSums(contagem) != 0)] #em != a exclamação inverte o sentido
zero_sum2 <- names(which(colSums(m_part_cols) == 0))
zero_sum2 #nomes das espécies zeradas
sum<-colSums(m_part_cols)
sum

###LINHAS ZERADAS DE CONTAGEM----
m_part2 <- as.data.frame(t(m_part_cols))
sum <- colSums(m_part2)
sum
zero_sum <- names(which(colSums(m_part2) == 0))
zero_sum #nomes das espécies zeradas
m_part_rows <- m_part2[(colSums(m_part2) != 0)] #em != a exclamação inverte o sentido
zero_sum2 <- names(which(colSums(m_part_rows) == 0))
zero_sum2 #nomes das espécies zeradas
sum<-colSums(m_part_rows)
sum
m_cont <- (t(m_part_rows))
m_cont
sum(m_cont)
ncol(m_cont)

##SALVANDO MATRIZ FINAL DE CONTAGEM----

write.table(m_cont, "res_ms/m_cont.csv",
```

```
            sep = ";", dec = ".", #"\t",
            row.names = TRUE,
            quote = TRUE,
            append = FALSE)
m_cont <- read.csv("res_ms/m_cont.csv",
                   sep = ";", dec = ".",
                   row.names = 1,
                   header = TRUE,
                   na.strings = NA)
m_cont

#ZOOREBIO23_PART - ORGANIZANDO ESCALAS DA COMUNIDADE----
#####....----

dev.off() #apaga os graficos, se houver algum
rm(list=ls(all=TRUE)) #limpa a memória
cat("\014") #limpa o console

library(openxlsx)

##CARREGANDO MATRIZES BRUTAS----

escalas <- read.xlsx("dados/rebio23-zoopl.xlsx",
                   rowNames = T,
                   colNames = T,
                   sheet = "escalas_final",
                   startRow = 2)

rownames(escalas)
del_rows <- c("C-C-P11-H1", "C-C-P11-H2", "C-C-P11-H3", "S-C-P11-H1", "S-C-P11-H2", "S-C-P11
del_rows
escalas <- escalas[!(row.names(escalas) %in% c(del_rows)),]

write.table(escalas, "res_ms/m_esc.csv",
            sep = ";", dec = ".", #"\t",
            row.names = TRUE,
            quote = TRUE,
            append = FALSE)
m_esc <- read.csv("res_ms/m_esc.csv",
                   sep = ";", dec = ".",
                   row.names = 1,
                   header = TRUE,
                   na.strings = NA)
```

```r
m_esc

#REBIO23-part----

##ORGANIZANDO DADOS----

dev.off()
rm(list=ls(all=TRUE))
cat("\014")

t_grps <- read.csv("res_ms/t_grps.csv",
                   sep = ";", dec = ".",
                   row.names = 1,
                   header = TRUE,
                   na.strings = NA)
m_hab <- read.csv("res_ms/m_hab_part.csv",
                  sep = ";", dec = ".",
                  row.names = 1,
                  header = TRUE,
                  na.strings = NA)

##CORRELOGRAMA E REMOÇÃO DE VARIÁVEIS REDUNDANTES OU DESNECESSÁRIAS----

library(psych)
colnames(m_hab)
#png("res_ms/fig-h.hab_pairs.png")
pairs.panels(m_hab[,1:8],
             method = "pearson", # correlation method
             scale = FALSE, lm = FALSE,
             hist.col = "#00AFBB", pch = 19,
             density = TRUE,  # show density plots
             ellipses = TRUE, # show correlation ellipses
             alpha = 0.5)
#dev.off()

cor <- cor(m_hab)
cor

library(corrplot)
#png("res_ms/fig-hab_corrplot.png")
corrplot(cor, method = "circle")
#dev.off()

```

```r
##DELETANDO COLINEARES OU INDESEJADAS----

#sink(file = "res_ms/colineares.txt", append = F, split = T)
colnames(m_hab)
del_cols <- c("", "")
#del_cols <- grep("^n\\.", colnames(m_hab), value = TRUE)
m_hab_part <- m_hab[, !(colnames(m_hab) %in% del_cols)]

##SOMANDO REDUNDANTES----

m_hab_part$h.Algae <- m_hab_part$h.Algae_f + m_hab_part$h.Algae_a
m_hab_part <- m_hab_part[, !(colnames(m_hab_part)
                             %in% c("h.Algae_f", "h.Algae_a"))]
m_hab_part$h.Debris <- m_hab_part$h.Debris_l + m_hab_part$h.Debris_s
m_hab_part <- m_hab_part[, !(colnames(m_hab_part)
                            %in% c("h.Debris_l", "h.Debris_s"))]

colnames(m_hab_part)
m_hab_part
#sink()

write.table(m_hab_part, "res_ms/m_hab_part.csv",
            sep = ";", dec = ".", #"\t",
            row.names = TRUE,
            quote = TRUE,
            append = FALSE)
m_hab_part <- read.csv("res_ms/m_hab_part.csv",
                       sep = ";", dec = ".",
                       row.names = 1,
                       header = TRUE,
                       na.strings = NA)
```

```r
colnames(m_hab_part)
library(PerformanceAnalytics)
chart.Correlation(m_hab_part[, c(1:20)], histogram=TRUE)
# ***p<0.001, **p<0.01, *p<0.05, .p<0.10

library(Hmisc)
mat <- as.matrix(m_hab_part[, 1:20])
res <- rcorr(mat, type = "pearson")
cor_mat <- res$r
p_mat   <- res$P

```

```r
stars <- ifelse(p_mat < 0.001, "***",
          ifelse(p_mat < 0.01,  "**",
          ifelse(p_mat < 0.05,  "*", "")))

cor_star <- ifelse(p_mat < 0.05,
                   paste0(round(cor_mat, 2), stars),
                   NA)

cor_star <- matrix(
  cor_star,
  nrow = nrow(cor_mat),
  dimnames = dimnames(cor_mat)
)

cor_star[upper.tri(cor_star, diag = TRUE)] <- NA
cor_star <- noquote(cor_star)
print(cor_star, na.print = "")
```

```r
#PCA fviz package----
#browseURL("https://www.sthda.com/english/articles/31-principal-component-methods-in-r-pract

#ORGANIZANDO DADOS----

dev.off()
rm(list=ls(all=TRUE))
cat("\014")

t_grps <- read.csv("res_ms/t_grps.csv",
                   sep = ";", dec = ".",
                   row.names = 1,
                   header = TRUE,
                   na.strings = NA)
m_hab_part <- read.csv("res_ms/m_hab_part.csv",
                       sep = ";", dec = ".",
                       row.names = 1,
                       header = TRUE,
                       na.strings = NA)

colnames(m_hab_part)

###RELATIVIZAÇÕES E TRANSFORMAÇÕES----
library(vegan)
library(tidyverse)
```

```r

# Selecionando variáveis que começam com "w" ou "m"
m_wm <- m_hab_part[, grep("^(w|m)", colnames(m_hab_part))]

# Selecionando variáveis que começam com "s" ou "h"
m_sh <- m_hab_part[, grep("^(s|h)", colnames(m_hab_part))]

# Removendo os dois primeiros caracteres dos nomes das colunas
colnames(m_wm) <- sub("^..", "", colnames(m_wm))
colnames(m_sh) <- sub("^..", "", colnames(m_sh))

# Transformações
m_wm <- sqrt(m_wm)
m_sh <- asin(sqrt(decostand(m_sh, method = "total", MARGIN = 2)))

# Juntando tudo em uma única tabela
hab_trns <- cbind(m_sh, m_wm)

# Salvado m_hab_trns
write.table(hab_trns, "res_ms/m_hab_trns.csv",
            sep = ";", dec = ".", #"\t",
            row.names = TRUE,
            quote = TRUE,
            append = FALSE)
m_hab_trns <- read.table("res_ms/m_hab_trns.csv",
                         sep = ";", dec = ".",
                         row.names = 1,
                         header = TRUE,
                         na.strings = NA)
```

These data were then analyzed using Principal Component Analysis (PCA) to summarize multivariate patterns of environmental variation among sampling sites. Prior to PCA, site morphology and water-quality variables were square-root transformed (Sokal and Rohlf 1995). Variables describing sediment composition and marginal habitat structure, expressed as percentages, were relativized by their column totals and subsequently subjected to the arcsine square-root transformation (McCune and Grace 2002). PCA was conducted in R using the `FactoMineR` package (Lê et al. 2008), with all variables centered and standardized to unit variance before analysis. The water quality parameters temperature and dissolved oxygen, and all nutrients were used to characterize differences between streams but were not used in the PCA because they were not measured for both seasons.A Permutational Multivariate Analysis of Variance (PERMANOVA, 999 permutations) was performed to test whether the overall set of environmental variables measured differed across the seasons and streams .

To compare the diversity of zooplankton taxa between streams, the Chao1 estimated richness (± SE) and the curves of rarefied and extrapolated richness were generated (95% upper and lower confidence intervals) using individual-based rarefaction curves (Gotelli and Colwell 2001, Freiry et al. 2020). Community composition and environmental variables were compared among hydrological periods and streams using Permutational Multivariate Analysis of Variance (PERMANOVA), based on a Bray–Curtis dissimilarity matrix (Anderson and Walsh 2013, Freiry et al. 2020). The model included the fixed factors Season, Stream and their interaction. The significance of the terms was assessed using 999 random permutations and the *pseudo*-F statistic (Anderson, 2001). Homogeneity of multivariate dispersion among groups was assessed based on the distance of each sampling unit from the centroid of its respective group in multivariate space, using the *betadisper* function from the `vegan` package (Anderson et al., 2006).

Differences in assemblage composition across the spatial hierarchy and seasons were explored using the Unweighted Pair Group Method with Averages (UPGMA) (Belbin and McDonald 1993, Ávila et al. 2011), based on Bray-Curtis dissimilarities calculated from the zooplankton density matrix (Ávila et al. 2011). Prior to the analyses, density data were relativized and square-root transformed to reduce the influence of highly abundant taxa and improve the distributional properties of the dataset (Rocha et al. 2012). To focus on spatial variation across sampling sites, habitat density data were pooled for this analysis. To assess how well the UPGMA dendrograms represented the original Bray-Curtis dissimilarities among sampling sites, the Cophenetic Correlation Coefficient (CC) was calculated (Sokal and Rohlf 1962, Sneath and Sokal 1973, Saraçli et al. 2013), where a cophenetic matrix was constructed from each seasonal dendrogram using the *cophenetic* function in the `vegan` package in R (Oksanen et al. 2020) . The cophenetic distance matrix was then correlated with the corresponding original Bray-Curtis dissimilarity matrix using Pearson's product-moment correlation coefficient (Sokal and Rohlf 1962, Farris 1969, Silva 2013). To evaluate the strength of the hierarchical clustering structure obtained from the UPGMA dendrograms, the Agglomerative Coefficient (AC) was calculated for each season using the *coef.hclust* function from the `cluster` package in R (Kaufman and Rousseeuw 1990, Maechler et al. 2026). The AC ranges from 0 to 1, where values closer to 1 indicate a strong, well-defined clustering structure and values near 0 reflect weak or indistinct grouping.

Additive diversity partitioning was used to quantify the contribution of within- and between-unit diversity to total zooplankton diversity. In this framework, alpha diversity (α) represents diversity occurring within sampling units, whereas beta diversity (β) represents the contribution of differences among units at successive levels of the spatial hierarchy (Lande 1996, Crist et al. 2003). The hierarchical structure included three spatial scales. The finest scales consisted of the habitat samples within each site (n = 3), and of each site (n = 6, for the Barro Branco stream, and n = 9, for the Caiana stream). The intermediate scales corresponded to the upper and lower stream sections (n = 2), whereas the broadest scale represented differences among the two streams. Accordingly, the total diversity (γ) was decomposed into within-habitat diversity (α_1) and the successive beta components associated with differences among habitats (β_1), among sites withing upper and lower stream reaches (β_2), and among streams (β_3). The contribution

of each component was expressed as a percentage of total diversity (Crist et al. 2003, Farias et al. 2020). To investigate seasonal variation in the hierarchical structure of diversity, additive partitioning was performed separately for each season and for each stream (Melo and Medeiros 2013, Farias et al. 2020).

The overall diversity was taken to be the sum of the relative contributions of each scale in the hierarchical sampling design. Analysis of significance of the contribution of each hierarchical level was performed through with individual-based randomizations (type I null model) (see Freiry et al. 2020, Melo and Medeiros 2013). The randomization process was repeated 999 times to obtain null distributions of the α and β diversity at each hierarchical level, and used to test whether the observed values were higher or lower than expected by chance (Crist et al. 2003). The probabilities obtained from randomization tests were interpreted as P-values, as a low P-value ($\text{Prop}_{exp} > \text{Prop}_{obs} < 0.05$) indicates an observed diversity significantly different than that expected under the null model. On the other hand, a high P-value ($\text{Prop}_{exp} > \text{Prop}_{obs} > 0.95$) indicates an observed diversity not different than that expected by chance (Veech et al. 2002, Gering et al. 2003, Crist et al. 2003, Freiry et al. 2020). The results of the analysis were subject to a Monte Carlo permutation test (999 permutations) for validation. The additive partitioning of the diversity was carried out with the function *adipart* from package `vegan` (Oksanen et al. 2020). All analyses were conducted in the statistical environment R (R Development Core Team 2017).

3 Results

Water velocity ranged from 0 to 0.41 m/s, with the highest average recorded at site 14. Waters were, on average, well oxygenated (5.3 to 8.1 mg/L), and cool, with average temperatures betwen from 25.2 to 32.5 °C, the average being 27.7 °C. Conductivity ranged from 98.8 and 223.0 S/cm, pH was mostly neutral to slightly acidic (from 5.1 to 6.9) and turbidity was low (4.9 to 79.2 NTU). Regarding the morphologacal variables, elevation ranged from 41.5 to 112.8 m a.s.l., while the stream margins slope reached 50° for some sites. Stream width and depth showed greater variations reflecting the range of habitats along the study streams, from small and narrow streams to wider and deeper manmade reservoirs in the stream channel. Stream site width varied from 1 m across to 19.5 m in larger reservoirs. Overall, depths ranged from approx. 10-20 cm to a maximum of 1.55 m, on average. Substrate composition across sites consisted mainly of mud (reaching 100% at sites 1 and 2, in the Barro Branco stream) and sand being more common at the Caiana Stream (reaching 98.3% cover at site 14) (Table 1). Marginal and underwater habitat structures were diverse with most variables being common at the Barro Branco stream, namely aquatic macrophytes, fillamentous and attached algae and littoral grass. Submerged and overhanging vegetation, leaf litter, root masses and wood debris were present at all sites (see Table 1).

```
#TABELA DE HABITAT----

```

```r
dev.off()
rm(list=ls(all=TRUE))
cat("\014")

habitat <- read.xlsx("dados/rebio23-habitat.xlsx",
                     rowNames = T,
                     colNames = T,
                     sheet = "ambiente",
                     startRow = 2)
m_hab_part <- read.csv("res_ms/m_hab_part.csv",
                       sep = ";", dec = ".",
                       row.names = 1,
                       header = TRUE,
                       na.strings = NA)
t_grps <- read.csv("res_ms/t_grps.csv",
                   sep = ";", dec = ".",
                   row.names = 1,
                   header = TRUE,
                   na.strings = NA)

m_hab_part$w.Temp_C <- habitat[rownames(m_hab_part), "w.Temp_C"]
m_hab_part$w.DO_mg.L <- habitat[rownames(m_hab_part), "w.DO_mg.L"]

library(dplyr)
library(tidyr)
m_trab <- m_hab_part %>%
  rename_with(~ gsub("_", ".", .))
m <- m_trab %>%
  group_by(PontoN = t_grps$PontoN) %>%
  summarise(across(where(is.numeric),
                   list(mean = ~ mean(.x, na.rm = TRUE), #ajustes celulas vazias
                        min = ~ min(.x, na.rm = TRUE),
                        max = ~ max(.x, na.rm = TRUE))),
            .groups = 'drop') %>%
  pivot_longer(
    cols = -c(PontoN),
    names_to = c("Variable", ".value"),
    names_sep = "_"
  )

m <- as.data.frame(m)
m_wide <- m %>%
  mutate(stat_string = ifelse(Variable == c("m.Vel.m.s"),
```

```
                                paste0(round(mean, 3), " (", round(min, 3), "-", round(max, 3)
                                paste0(round(mean, 1), " (", round(min, 1), "-", round(max, 1)
  unite("Location", PontoN, sep = "_") %>%
  select(Variable, Location, stat_string) %>%
  pivot_wider(names_from = Location, values_from = stat_string)

m_wide
m_wide <- as.data.frame(m_wide)
cols <- colnames(m_wide)
col_order <- c(
  "Variable",
  cols[cols != "Variable"][
    order(as.numeric(gsub("Ponto", "", cols[cols != "Variable"])))
  ]
)
m_wide <- m_wide[, c(col_order)] #sequência das colunas
row_order <- c(
  "w.Temp.C",
  "w.DO.mg.L",
  "w.Vel.m.s",
  "w.Cond.uS.cm",
  "w.Turb.ntu",
  "w.pH",
  "m.Elev.m",
  "m.Slope",
  "m.Depth.avg.cm",
  "m.Depth.mar.cm",
  "m.Depth.max.cm",
  "m.Width.m",
  "s.Mud",
  "s.Sand",
  "s.Gravel.f",
  "s.Rocks",
  "h.Macrophytes",
  "h.Algae",
  "h.Subm.veg",
  "h.Overh.veg",
  "h.Grass",
  "h.Leaf.l",
  "h.Root.masses",
  "h.Debris"
)
m_wide <- m_wide[match(row_order, m_wide$Variable), ]
```

```r
m_wide

# Salvado m_wide
write.table(m_wide, "res_ms/m_wide_hab_to.txt",
            sep = ";", dec = ".", #"\t",
            row.names = TRUE,
            quote = TRUE,
            append = FALSE)
m_wide_hab_to <- read.table("res_ms/m_wide_hab_to.txt",
                  sep = ";", dec = ".",
                  row.names = 1,
                  header = TRUE,
                  na.strings = NA)

# Exportando dados para Excel
library(openxlsx)
#write.xlsx(m_wide, file = "res_ms/tabela de habitat.xlsx", rowNames = FALSE)
wb <- loadWorkbook("res_ms/tabela de habitat.xlsx")
writeData(wb, sheet = "Sheet 1", x = m_wide)
saveWorkbook(wb, "res_ms/tabela de habitat.xlsx", overwrite = TRUE)

# Tabela final ajustada no Excel
library(readxl)
hab_gttable <- read_excel(
  path  = "res_ms/tabela de habitat.xlsx",
  sheet = "tabela_final")
#range = "A1:P30"#ajustar para o tamanho da tabela final
hab_gttable
library(gt)
hab_gttable <- gt(hab_gttable)
hab_gttable <- sub_missing(hab_gttable,
                      columns = everything(), missing_text = "")
#hab_gttable <- fmt_number(hab_gttable, columns = "F.O.(%)", decimals = 1)
hab_gttable
gtsave(hab_gttable, "res_ms/gt-hab_gttable_xlsx.html")
saveRDS(hab_gttable, "res_ms/gt-hab_gttable_xlsx.rds")
```

```r
#TABELA DE HABITAT----
##SEGUE DO SCRIPT ANTERIOR

# Escolher sumário de uma variavel
m
var <- "w.Vel.m.s"
```

```r
m[m$Variable == var, "mean"] #cada valor de var
summary(m[m$Variable == var, "mean"]) #sumário dos valores de var

# Escolher sumário de um grupo de variáveis do df m
vars <- unique(grep("^h\\.", m$Variable, value = TRUE))
summaries <- list() #criam uma lista vazia para guardar os sumários
# Loop para cada variável do grupo e guarda em summaries
for (var in vars) {
  summaries[[var]] <- summary(m[m$Variable == var, "mean"])
}
#var is a temporary variable used in the for loop to iterate through
#each variable name that starts with "h."

summaries
summary_table <- do.call(rbind, lapply(summaries, as.data.frame.list))
colnames(summary_table) <- c("Min", "Q1", "Median", "Mean", "Q3", "Max")
#sink(file = "res_ms/summary_w.txt", split = TRUE)
round(summary_table, 2)
#round(summary_table[order(summary_table$Mean, decreasing = FALSE), ], 2)
#sink()

#Summario para as médias

dev.off()
rm(list=ls(all=TRUE))
cat("\014")

m_hab_part <- read.csv("res_ms/m_hab_part.csv",
                       sep = ";", dec = ".",
                       row.names = 1,
                       header = TRUE,
                       na.strings = NA)
t_grps <- read.csv("res_ms/t_grps.csv",
                   sep = ";", dec = ".",
                   row.names = 1,
                   header = TRUE,
                   na.strings = NA)

library(dplyr)
library(tidyr)

m_trab <- m_hab_part %>%
  rename_with(~ gsub("_", ".", .))
```

```r

# Compute averages per group
m_s <- m_trab %>%
  group_by(PontoN = t_grps$PontoN) %>%
  summarise(across(where(is.numeric), ~ mean(.x, na.rm = TRUE)),
            .groups = 'drop') %>%
  pivot_longer(
    cols = -PontoN,
    names_to = "Variable",
    values_to = "mean"
  )

# Format rounding and reshape to wide format
m_wide_s <- m_s %>%
  mutate(
    stat_string = ifelse(
      Variable == "m.Vel.m.s",
      as.character(round(mean, 3)),
      as.character(round(mean, 1))
    )
  ) %>%
  unite("Location", PontoN, sep = "_") %>%
  select(Variable, Location, stat_string) %>%
  pivot_wider(names_from = Location, values_from = stat_string)

m_wide_s <- as.data.frame(m_wide_s)

# Order columns numerically by Ponto number
cols <- colnames(m_wide_s)
col_order <- c(
  "Variable",
  cols[cols != "Variable"][
    order(as.numeric(gsub("Ponto", "", cols[cols != "Variable"])))
  ]
)
m_wide_s <- m_wide_s[, col_order]

# Order rows according to pre-defined variable list
row_order <- c(
  "w.Cond.uS.cm",
  "w.Turb.ntu",
  "w.pH",
  "m.Elev.m",
```

```
  "m.Vel.m.s",
  "m.Slope",
  "m.Depth.avg.cm",
  "m.Depth.mar.cm",
  "s.Mud",
  "s.Sand",
  "s.Gravel.f",
  "s.Rocks",
  "h.Macrophytes",
  "h.Algae",
  "h.Subm.veg",
  "h.Overh.veg",
  "h.Grass",
  "h.Leaf.l",
  "h.Root.masses",
  "h.Debris"
)

m_wide_s <- m_wide_s[match(row_order, m_wide_s$Variable), ]

# View final table
m_wide_s
```

Principal Component Analysis (PCA) was used to summarize environmental variation among sampling sites and to identify variables contributing to the differences between streams (Figure 2) and seasons (Figure 3). The first two principal components explained 38.6% of the total variance, with the first axis accounting for 23.2%. In relation to the environmental variables of water quality, the Barro Branco stream was more associated with higher elevations, the predominance of muddy substrate and a narrower array of underwater structures, mostly algae, macrophytes, leaf litter and woody debris. The Caiana stream showed a wider range of variables affecting it's waters, being associated with higher pH, turbidity and depths, as well as a predominance of sandy substrates and, to a lesser degree, root masses. Water velocity was also important for this stream. PERMANOVA showed significant differences in the environmental variables between the two study streams ($pseudo\text{-}\mathrm{F}_{1,86} = 8.3, R^2 = 0.082, p = 0.001$). Even though dry and wet seasons did not show clear segregation in the PCA plot, PERMANOVA indicated significance of differences ($pseudo\text{-}\mathrm{F}_{1,86} = 4.2, R^2 = 0.042, p = 0.006$). A significant Season $\times$ Stream interaction was also detected ($pseudo\text{-}\mathrm{F}_{1,86} = 2.8, R^2 = 0.028, p = 0.032$), indicating that the effect of hydrological period on the environmental variables differed between Barro Branco and Caiana. The complete model explained 15.1% of the variation, whereas 84.9% remained unexplained by the factors included in the model.

```
#PCA fviz package----
#browseURL("https://www.sthda.com/english/articles/31-principal-component-methods-in-r-pract
```

```r

#ORGANIZANDO DADOS----

dev.off()
rm(list=ls(all=TRUE))
cat("\014")

t_grps <- read.csv("res_ms/t_grps.csv",
                   sep = ";", dec = ".",
                   row.names = 1,
                   header = TRUE,
                   na.strings = NA)
m_hab_trns <- read.csv("res_ms/m_hab_trns.csv",
                       sep = ";", dec = ".",
                       row.names = 1,
                       header = TRUE,
                       na.strings = NA)

#PCA----

library("FactoMineR")
library("factoextra")

pca <- PCA(m_hab_trns, scale.unit = TRUE, ncp = 5, graph = TRUE)
print(pca)

eig.val <- get_eigenvalue(pca)
eig.val

#sink(file = "res_ms/cor_matrix.txt", append = FALSE)
pca$var$cor
#sink()

fviz_eig(pca, addlabels = TRUE, ylim = c(0,30))

var <- get_pca_var(pca)
var

fviz_pca_var(pca, col.var = "black")

library("corrplot")
corrplot(var$cos2, is.corr=FALSE)

```

```r
fviz_cos2(pca, choice = "var", axes = 1:2)

fviz_pca_var(pca, col.var = "cos2",
             gradient.cols = c("#00AFBB", "#E7B800", "#FC4E07"),
             repel = TRUE # Avoid text overlapping
)

fviz_pca_var(pca, alpha.var = "cos2")

corrplot(var$contrib, is.corr=FALSE)

fviz_contrib(pca, choice = "var", axes = 1, top = 10)
fviz_contrib(pca, choice = "var", axes = 2, top = 10)
fviz_contrib(pca, choice = "var", axes = 1:2, top = 10)

fviz_pca_var(pca, col.var = "contrib",
             gradient.cols = c("#00AFBB", "#E7B800", "#FC4E07")
)
fviz_pca_var(pca, alpha.var = "contrib")

##GROUPING BY KMEANS----

set.seed(123)
res.km <- kmeans(var$coord, centers = 3, nstart = 25)
grp <- as.factor(res.km$cluster)
# Color variables by groups
fviz_pca_var(pca, col.var = grp,
             palette = c("#0073C2FF", "#EFC000FF", "#868686FF"),
             legend.title = "Cluster")

ind <- get_pca_ind(pca)
ind
ind$contrib

##BIPLOTS----

fviz_pca_ind(pca)

fviz_pca_ind(pca, col.ind = "cos2",
             gradient.cols = c("#00AFBB", "#E7B800", "#FC4E07"),
             repel = TRUE #avoid text overlapping (slow if many points)
)

```

```r
fviz_pca_ind(pca, pointsize = "cos2",
             pointshape = 21, fill = "#E7B800",
             repel = TRUE #avoid text overlapping (slow if many points)
)

fviz_pca_ind(pca, col.ind = "cos2", pointsize = "cos2",
             gradient.cols = c("#00AFBB", "#E7B800", "#FC4E07"),
             repel = TRUE #avoid text overlapping (slow if many points)
)

fviz_cos2(pca, choice = "ind")

fviz_contrib(pca, choice = "ind", axes = 1:2)

# Create a random continuous variable of length 23,
# Same length as the number of active individuals in the PCA
t_grps <- t_grps[rownames(pca$ind$coord), ]
colnames(t_grps)
set.seed(123)
# Escolhendo variável categorica
my.cont.var <- t_grps$Periodo
my.cont.var[my.cont.var == "seco"] <- "dry"
my.cont.var[my.cont.var == "chuvoso"] <- "wet"
# Color individuals by the continuous variable
fviz_pca_ind(
  pca,
  habillage = my.cont.var,
  palette = c("blue", "yellow", "red", "pink"),
  repel = FALSE
)

fviz_pca_ind(pca,
             geom.ind = "point", #show points only (nbut not "text")
             col.ind = as.factor(my.cont.var), #color by groups
             palette = "grey",
             addEllipses = TRUE, #concentration ellipses
             legend.title = "Groups"
)

length(unique(my.cont.var))
fviz_pca_ind(pca,
             geom.ind = "point",
             col.ind = my.cont.var,
```

```
              palette = "grey",
              addEllipses = TRUE, ellipse.type = "confidence",
              mean.point = TRUE,
              legend.title = "Groups")

fviz_pca_biplot(pca, repel = TRUE,
                col.var = "#2E9FDF", #variables color
                col.ind = "#696969"  #individuals color
)

# PCA com elipse colorida
fviz_pca_biplot(pca,
                col.ind = my.cont.var, palette = "jco",
                addEllipses = TRUE, ellipse.type = "t", #t-distribution confidence ellipses"
                ellipse.level = 0.95,
                shape.ind = my.cont.var,
                label = "var",
                col.var = "black", repel = TRUE,
                legend.title = "Stream") +
  ggtitle(NULL) +
  scale_color_manual(values = c(
    "#1B9E77",
    "#D95F02",
    "#7570B3",
    "#E7298A"
  )) +
  scale_shape_manual(values = c(
    16, # circle
    17, # triangle
    15, # square
    18  # diamond
  ))

### FILTRANDO AS VARIÁVEIS
var_coord <- pca$var$coord[,1:2]
keep_vars <- apply(abs(var_coord) > 0.2,1,any)
vars_to_keep <- rownames(var_coord)[keep_vars]
pca_filt <- pca
pca_filt$var$coord    <- pca$var$coord[vars_to_keep,,drop=FALSE]
pca_filt$var$cos2     <- pca$var$cos2[vars_to_keep,,drop=FALSE]
pca_filt$var$contrib  <- pca$var$contrib[vars_to_keep,,drop=FALSE]

### AJUSTE MANUAL DAS COORDENADAS DOS VETORES
```

```r

# Cria uma tabela SEPARADA apenas para as posições dos rótulos
label_coords <- as.data.frame(pca_filt$var$coord)
# Multiplica APENAS a posição do texto (as setas continuam intactas no pca_filt)
label_coords <- label_coords * 2.0
label_coords <- label_coords * 2.0
# Se quiser ajustar pontos específicos manualmente
#fix(label_coords) #nomes das colunas
#write.table(label_coords,"res_ms/coords_lab.txt")
label_coords <- read.table("res_ms/coords_lab.txt",
                           header=TRUE,
                           row.names=1)

# GRÁFICO FINAL

#png("res_ms/fig-hab_PCA_periodo.png", width = 9, height = 7.5, units = "in", res = 300)
fviz_pca_biplot(pca_filt,
  col.ind = my.cont.var, palette = "jco",
  addEllipses = TRUE, ellipse.type = "t",
  ellipse.level = 0.95,
  shape.ind = my.cont.var,
  label = "none",
  col.var = "black",
  legend.title = "Season"
) +
  # Adiciona os rótulos nas posições customizadas da tabela label_coords
  geom_text(
    data = label_coords,
    aes(x = Dim.1, y = Dim.2, label = rownames(label_coords)),
    color = "black",
    size = 12) +
  ggtitle(NULL) +
  scale_color_manual(values = c("#1B9E77", "#D95F02", "#7570B3", "#E7298A")) +
  scale_fill_manual(values = c("darkcyan", "orange", "#7570B3", "#E7298A")) +
  scale_shape_manual(values = c(
    16, # circle
    17, # triangle
    15, # square
    18  # diamond
  )) +
  theme(
  axis.title = element_text(size = 34),
  axis.text = element_text(size = 32),
```

```
  legend.title = element_text(size = 40),
  legend.text = element_text(size = 40)
)
#dev.off()
```

```
#ORGANIZANDO DADOS----

dev.off()
rm(list=ls(all=TRUE))
cat("\014")

t_grps <- read.csv("res_ms/t_grps.csv",
                   sep = ";", dec = ".",
                   row.names = 1,
                   header = TRUE,
                   na.strings = NA)
m_hab_trns <- read.csv("res_ms/m_hab_trns.csv",
                       sep = ";", dec = ".",
                       row.names = 1,
                       header = TRUE,
                       na.strings = NA)

colnames(t_grps)
m_trab <- m_hab_trns
library(vegan)

#MRPP----

#TRABALHANDO COM MULTIPLAS VARIAVEIS----

# Run MRPP for multiple variables
fatores <- c("Periodo", "Rio", "Ref_site")
set.seed(123)
mrpp_results <- lapply(fatores, function(v) {
  grp <- as.factor(t_grps[[v]])
  mrpp(
    dat = m_trab,
    grouping = grp,
    distance = "bray"
  )
})
names(mrpp_results) <- fatores
mrpp_results
```

```r
# Print one result
mrpp_results$Ref_site
# Loop through summaries
sink(file = "res_ms/mrpp-env.txt", split = TRUE)#, append = TRUE)
for (v in names(mrpp_results)) {
  cat("\n============================\n")
  cat("MRPP for:", v, "\n")
  print(mrpp_results[[v]])
}
sink()
# Extract key statistics into a table
mrpp_summary <- data.frame(
  Variable = fatores,
  Delta = sapply(mrpp_results, function(x) x$delta),
  Expected_Delta = sapply(mrpp_results, function(x) x$E.delta),
  A = sapply(mrpp_results, function(x) x$A),
  P_value = sapply(mrpp_results, function(x) x$Pvalue)
)
sink("res_ms/mrpp-env.txt", append = TRUE)
mrpp_summary
sink()

##TRABALHANDO COM UMA UNICA VARIAVEL----

# Run MRPP for a single variable
grp <- as.factor(t_grps$Ref_site) #UA, mes, estacao, area, ambiente
set.seed(123)
mrpp <- mrpp(dat = m_trab, grouping = grp, distance = "bray")
mrpp
summary(mrpp)

# Teste de hipótese para uma unica variavel
str(mrpp)
p_value <- mrpp$Pvalue
print(sprintf("p-value: %.10f", p_value))
# Conditional statement to check the p-value and print the appropriate message
if (p_value < 0.05) {
  print(sprintf("Existe diferença significativa porquê o valor de p é %.10f, sendo menor que
} else {
  print(sprintf("Não existe diferença significativa porquê o valor de p é %.10f, sendo maior
}

##TESTES PAREADOS----
```

```r

# Todos os pares de níveis do fator
pairs <- combn(levels(as.factor(grp)), 2, simplify = FALSE)
pairs
# Aplicar mrpp para cada par
res_list <- lapply(pairs, function(p) {
  sel <- grp %in% p
  mrpp(dat = m_trab[sel, ], grouping = droplevels(grp[sel]), distance = "bray")
})
# Dar nomes aos resultados
names(res_list) <- sapply(pairs, function(p) paste(p, collapse = "_vs_"))
# visualizar
res_list
# Resumo
summary_tab <- data.frame(
  Comparison = names(res_list),
  A = sapply(res_list, function(x) x$A),
  Delta = sapply(res_list, function(x) x$delta),
  Pvalue = sapply(res_list, function(x) x$Pvalue)
)
sink("res_ms/mrpp-env.txt", append = TRUE)
summary_tab
sink()

###MATRIX-STYLE TABLE----
library(dplyr)
library(tidyr)
library(knitr)
library(kableExtra)
# Extract unique group names from the comparisons
groups1 <- sort(unique(unlist(strsplit(summary_tab$Comparison, "_vs_"))))
# Initialize empty matrix
pval_matrix <- matrix(NA, nrow = length(groups1), ncol = length(groups1),
                      dimnames = list(groups1, groups1))
# Fill matrix with Pvalues
for (i in 1:nrow(summary_tab)) {
  comp <- unlist(strsplit(summary_tab$Comparison[i], "_vs_"))
  g1 <- comp[1]
  g2 <- comp[2]
  p <- summary_tab$Pvalue[i]

  pval_matrix[g1, g2] <- p
  pval_matrix[g2, g1] <- p  #make symmetric
```

```r
}

# Format p-values: red if <0.05
pval_colored <- ifelse(is.na(pval_matrix), "",
                       ifelse(pval_matrix < 0.05,
                              paste0("<span style='color:red;'>", sprintf("%.3f", pval_matri
                              sprintf("%.3f", pval_matrix)))

# Convert to data.frame for kable
pval_df <- as.data.frame(pval_colored)
pval_df <- tibble::rownames_to_column(pval_df, "Group")
# Render HTML table
matrixstyle <- pval_df %>%
  kable("html", escape = FALSE) %>%
  kable_styling(full_width = FALSE)
matrixstyle
save_kable(matrixstyle, "res_ms/kb-mrpp-env_matrixstyle.html")

##GRÁFICO DA MRPP----

png("res_ms/fig-mrpp-env.png", width = 15, height = 10, units = "in", res = 300)
def.par <- par(no.readonly = TRUE)
layout(matrix(1:2,nr=1))
plot(ord <- metaMDS(m_trab, distance="bray"), type="text", display="sites")
with(t_grps, ordihull(ord, grp))
with(mrpp, {
  fig.dist <- hist(boot.deltas, xlim=range(c(delta,boot.deltas)),
                   main="Test of Differences Among Groups")
  abline(v=delta);
  text(delta, 2*mean(fig.dist$counts), adj = -0.5,
       expression(bold(delta)), cex=1.5 )  }
)
par(def.par)
dev.off()

###AVALIANDO AS DISTÂNCIAS MÉDIAS----
md <- with(t_grps, meandist(vegdist(m_trab), grp))
md
summary(md)
par(mfrow=c(1,2))
plot(md)
plot(md, kind="histogram")
par(mfrow=c(1,1))
```

```r

#PerMANOVA----

#browseURL("https://riffomonas.org/code_club/2021-03-18-vegan")
#browseURL("https://riffomonas.org/code_club/2021-03-22-reproducibility")

library(tidyverse)
library(ggtext)
library(ggplot2)
library(vegan)

## TESTES PARA VARIAVEIS COM MAIS DE DOIS FATORES

adonis2(m_trab ~ Ref_site, data = t_grps, method = "bray")
set.seed(123)
perm <- 999
all_test <- adonis2(m_trab ~ Ref_site, data = t_grps, method = "bray", permutations = perm)
all_test

## TESTES PARA VARIAVEIS COM DOIS FATORES

sink(file = "res_ms/perma-env.txt", split = TRUE)#, append = TRUE)
set.seed(123)
perm <- 999
adonis2(m_trab ~ Periodo, data = t_grps, method = "bray", permutations = perm)
adonis2(m_trab ~ Rio, data = t_grps, method = "bray", permutations = perm)
adonis2(m_trab ~ Periodo + Rio, data = t_grps, #+ no lugar de * tira os residuos
        method = "bray", permutations = perm)

## AVALIANDO INTERAÇÕES-----

adonis2(m_trab ~ Periodo*Rio, data = t_grps, method = "bray", permutations = perm,
        by = "terms") #ou by=NULL remove as interações
adonis2(m_trab ~ Periodo*Rio, data = t_grps, method = "bray", permutations = perm,
        by = "margin") #ou by=NULL remove as interações
adonis2(m_trab ~ Periodo*Rio, data = t_grps, method = "bray", permutations = perm,
        by = NULL) #ou by=NULL remove as interações
sink()

#BETADISPER

dist_bray <- vegdist(m_trab, method = "bray")
t_grps$Periodo_Rio <- interaction(
```

```r
  t_grps$Periodo,
  t_grps$Rio,
  sep = "_")
bd <- betadisper(dist_bray, t_grps$Periodo_Rio)
permutest(bd, permutations = 999)

# PARA CADA FATOR

bd_periodo <- betadisper(dist_bray, t_grps$Periodo)
bd_rio <- betadisper(dist_bray, t_grps$Rio)
permutest(bd_periodo, permutations = 999)
permutest(bd_rio, permutations = 999)

######### ARRUMAR ISSO ############################

all_p <- all_test$`Pr(>F)`[1]
all_p

count(t_grps, Periodo)
colnames(t_grps)

# Pairwise comparisons
#browseURL("https://doi.org/10.1007/s10641-018-0751-1")
#Lanés, L.E.K., Reichard, M., de Moura, R.G. et al. 2018.
#Environmental predictors for annual fish assemblages in subtropical
#grasslands of South America: the role of landscape and habitat
#characteristics. Environ Biol Fish 101, 963-977.

library(dplyr)
#library(combninat) # or base R combn()

set.seed(123)

# Get unique levels of UA
groups <- unique(t_grps$Ref_site)
groups
# All pairwise combinations
pairs <- combn(groups, 2, simplify = FALSE)
pairs
# Store results
pairwise_results <- lapply(pairs, function(g) {
  # Subset data
  sub_data <- t_grps %>% filter(Ref_site %in% g)
```

```r
  sub_dist <- m_trab[rownames(sub_data), ]

  # Run adonis2
  test <- adonis2(sub_dist ~ Ref_site, data = sub_data,
                  method = "bray", permutations = perm)

  tibble(
    group1 = g[1],
    group2 = g[2],
    F_value = test$F[1],
    R2 = test$R2[1],
    p_value = test$`Pr(>F)`[1]
  )
})
# Bind all results
pairwise_results <- bind_rows(pairwise_results)
pairwise_results
pairwise_p <- numeric()
pairwise_p <- pairwise_results$p_value
pairwise_p
adj_p_values <- p.adjust(pairwise_p, method="bonferroni") #c("holm", "hochberg", "hommel", "
adj_p_values
pairwise_results$adj_p_values <- adj_p_values
pairwise_results
# Get unique group names
groups2 <- sort(unique(c(pairwise_results$group1, pairwise_results$group2)))
# Initialize empty matrix
pval_matrix <- matrix(NA, nrow = length(groups2), ncol = length(groups2),
                      dimnames = list(groups2, groups2))
# Fill in p-values
for (i in 1:nrow(pairwise_results)) {
  g1 <- pairwise_results$group1[i]
  g2 <- pairwise_results$group2[i]
  p <- pairwise_results$p_value[i]

  pval_matrix[g1, g2] <- p
  pval_matrix[g2, g1] <- p  #make symmetric
}
pval_matrix

library(knitr)
library(kableExtra)

```

```r
# Make a copy of the matrix
pval_colored <- pval_matrix
# Replace numeric values with formatted HTML strings
pval_colored[] <- ifelse(is.na(pval_matrix), "",
                         ifelse(pval_matrix < 0.05,
                                paste0("<span style='color:red;'>", sprintf("%.3f", pval_mat
                                sprintf("%.3f", pval_matrix)))
# Convert to data frame with rownames
pval_df <- as.data.frame(pval_colored)
pval_df <- tibble::rownames_to_column(pval_df, "Group")
# Render table
pval_df %>%
  kable("html", escape = FALSE) %>%
  kable_styling(full_width = FALSE)
save_kable(matrixstyle, "res_ms/kb-perma_matrixstyle.html")

## AVALIANDO INTERAÇÕES-----

adonis2(m_trab ~ Periodo, data = t_grps, method = "bray")
adonis2(m_trab ~ Periodo*Rio, data = t_grps,
        method = "bray") #+ no lugar de * tira os residuos
adonis2(m_trab ~ Periodo*Rio, data = t_grps,
        method = "bray", by = "terms", strata = NULL)
```

Among the averaged nutrient concentrations (Table 2), ammonia ranged from 49.2 to 243.7 µg/L. The Caiana stream tended to have greater values, but overall no clear trend can be observed when comparing both streams. Nitrite and nitrate, on the other hand, show clearly a greater concentrations at the Caiana stream, nitrate varying from 166.4 to 2003.0 µg/L, compared with the maximum of 18.8 µg/L at the Barro Branco stream. Nitrite was too low to be detectable at the Barro Branco stream. The same pattern was observed for orthophosphate concentrations, being barely detectable at Barro Branco stream, and varying from 4.0 to 32.0 µg/L at the Caiana stream. Total phosphorus, however, tended to have higher concentrations at Barro Branco stream, reaching 50.1 µg/L at Site 3, but with highly variable values (from non detectable to 38.7 µg/L, as the second highest value).

```r
#TABELA DE NUTRIENTES----

##ORGANIZANDO DADOS----

dev.off()
rm(list=ls(all=TRUE))
cat("\014")

```

```r
t_nutri <- read.csv("res_ms/t_nutri.csv",
                    sep = ";", dec = ".",
                    row.names = 1,
                    header = TRUE,
                    na.strings = NA)

t_grps <- read.csv("res_ms/t_grps.csv",
                    sep = ";", dec = ".",
                    row.names = 1,
                    header = TRUE,
                    na.strings = NA)

nutri <- t_nutri

library(dplyr)
library(tidyr)
t_grps <- t_grps[seq_len(nrow(nutri)), , drop = FALSE]
m_trab <- nutri %>%
  rename_with(~ gsub("_", ".", .))
m <- m_trab %>%
  group_by(PontoN = t_grps$PontoN) %>%
  summarise(across(where(is.numeric),
                    list(mean = mean, min = min, max = max)),
            .groups = 'drop') %>%
  pivot_longer(
    cols = -c(PontoN),
    names_to = c("Variable", ".value"),
    names_sep = "_"
  )

m <- as.data.frame(m)
colnames(m_trab)
m_wide <- m %>%
  mutate(stat_string = ifelse(Variable == c("Ammonia.NH3.ug.L"),
                              paste0(round(mean, 1), " (", round(min, 1), "-", round(max, 1)
                              paste0(round(mean, 1), " (", round(min, 1), "-", round(max, 1)
  unite("Location", PontoN, sep = "_") %>%
  select(Variable, Location, stat_string) %>%
  pivot_wider(names_from = Location, values_from = stat_string)

m_wide
m_wide <- as.data.frame(m_wide)
cols <- colnames(m_wide)
```

```r
col_order <- c(
  "Variable",
  cols[cols != "Variable"][
    order(as.numeric(gsub("Ponto", "", cols[cols != "Variable"])))
  ]
)
m_wide <- m_wide[, c(col_order)] #sequência das colunas
m_wide

# Salvado m_wide
write.table(m_wide, "res_ms/m_wide_nutri.txt",
            sep = ";", dec = ".", #"\t",
            row.names = TRUE,
            quote = TRUE,
            append = FALSE)
m_wide_nutri <- read.table("res_ms/m_wide_nutri.txt",
                  sep = ";", dec = ".",
                  row.names = 1,
                  header = TRUE,
                  na.strings = NA)

# Exportando dados para Excel
library(openxlsx)
#write.xlsx(m_wide, file = "res_ms/tabela de nutrientes.xlsx", rowNames = FALSE)
wb <- loadWorkbook("res_ms/tabela de nutrientes.xlsx")
writeData(wb, sheet = "Sheet 1", x = m_wide_nutri)
saveWorkbook(wb, "res_ms/tabela de nutrientes.xlsx", overwrite = TRUE)

# Tabela final ajustada no Excel
library(readxl)
nutri_gttable <- read_excel(
  path  = "res_ms/tabela de nutrientes.xlsx",
  sheet = "tabela_final",
  range = "A1:P7") #ajustar para o tamanho da tabela final
nutri_gttable
library(gt)
nutri_gttable <- gt(nutri_gttable)
nutri_gttable <- sub_missing(nutri_gttable,
                      columns = everything(), missing_text = "")
nutri_gttable
gtsave(nutri_gttable, "res_ms/gt-nutri_gttable_xlsx.html")
saveRDS(nutri_gttable, "res_ms/gt-nutri_gttable_xlsx.rds")
```

A total of 56 taxa distributed among Rotifera (14 families), Cladocera (5 families) and Copepoda (2 families) were identified during the study period. Rotifera was the most species-rich group, represented by 39 species (plus Bdelloidea), followed by Cladocera (12 species) and Copepoda (4 species, in addition to the juvenile stages nauplii and copepodite). Across sampling periods, Lecanidae exhibited the highest richness among all recorded families of Rotifera with 15 species, followed by Lepadellidae, Synchaetidae (both with 4 species), Brachionidae and Mytilinidae (both with 3 species). Bdelloidea was the most abundant taxa (6194 individuals) followed by *Lecane curvicornis* with 626 individuals. Among the cladocerans, Chydoridae was the richest family with 5 species, followed by Macrothricidae (3 species). *Alonella dadayi* was the most abundant among the cladocerans with 1051 individuals. Juvenile stages (nauplii and copepodite) were also abundant with 2433 and 4061 individuals, respectively. Copepods were represented by he orders Cyclopoida and Calanoida, *Paracyclops* sp. *Mesocyclops* sp. being the most abundant (141 and 231 individuals, respectively) (Table 3).

```
#ZOOREBIO_part - ESTR. DA COMUNIDADE----
#####....----

##ORGANIZANDO DADOS----

dev.off() #apaga os graficos, se houver algum
rm(list=ls(all=TRUE)) #limpa a memória
cat("\014") #limpa o console

t_grps <- read.csv("res_ms/t_grps.csv",
                   sep = ";", dec = ".",
                   row.names = 1,
                   header = TRUE,
                   na.strings = NA)
m_dens <- read.csv("res_ms/m_dens.csv",
                   sep = ";", dec = ".",
                   row.names = 1,
                   header = TRUE,
                   na.strings = NA)
m_hab_part <- read.csv("res_ms/m_hab_part.csv",
                  sep = ";", dec = ".",
                  row.names = 1,
                  header = TRUE,
                  na.strings = NA)
m_cont <- read.csv("res_ms/m_cont.csv",
                   sep = ";", dec = ".",
                   row.names = 1,
                   header = TRUE,
                   na.strings = NA)
```

```r

library(tibble)
library(tidyverse)
library(forcats)
library(openxlsx)
library(Rcpp)

m_dens
str(m_dens)
m_trab <- m_dens

m_trab <- (m_dens) #SITES
m_trab <- t(m_dens) #SPP

##TAMANHO DA MATRIZ----

range(m_trab) #menor e maior valores
length(m_trab) #no. de colunas
ncol(m_trab) #no. de N colunas
nrow(m_trab) #no. de M linhas
sum(lengths(m_trab)) #soma os nos. de colunas
length(as.matrix(m_trab)) #tamanho da matriz m x n
sum(m_trab == 0) #número de observações igual a zero
sum(m_trab > 0) #número de observações maiores que zero
#calculando a proporção de zeros na matriz
zeros <- (sum(m_trab == 0)/length(as.matrix(m_trab)))*100
zeros

##ESTRUTURA DA COMUNIDADE----

Sum <- rowSums(m_trab)
#ou
Sum <- apply(m_trab,1,sum)
Sum
# Abundância relativa (%)
RA <- (Sum / sum(Sum)) * 100 # percentage
# Media
Mean <- rowMeans(m_trab)
Mean
# Ou
Mean <- apply(m_trab,1,mean)
Mean
# Desvio padrão
```

```r
DP <- apply(m_trab,1,sd)
DP
# Máximo
Max <- apply(m_trab,1,max)
Max
# Mínimo
Min <- apply(m_trab,1,min)
Min
# Mínimo não-zero
MinZ <- apply(m_trab, 1, function(row) {
  non_zero_values <- row[row > 0]  # Filter out zero values
  if (length(non_zero_values) == 0) {
    return(0)  # If all values are zero, return 0
  } else {
    return(min(non_zero_values))  # Return the minimum of non-zero values
  }
})
MinZ

m_pa <- m_trab
m_pa[m_pa != 0] <- 1
rowSums(m_pa)
library(vegan)
bin <- decostand(m_trab,"pa")
bin[1:10, 1:10]
S <- apply(bin,1,sum)
S
#OU
Riqueza <- specnumber(m_trab)
Riqueza
Riqueza_total <- specnumber(colSums(m_trab))
Riqueza_total

FO <- rowSums(m_trab > 0) / ncol(m_trab) * 100
FO

H <- diversity(m_trab, index = "shannon")
H
D <- diversity(m_trab, "simpson")
D
D[is.na(D)] <- 0 #substitui NA ou NaN por 0
D
E <- H/log(specnumber(m_trab))
```

```r
E
E[is.na(E)] <- 0 #substitui NA ou NaN por 0
E

###DESCRITORES----
Descritores1 <- cbind(Sum, RA, Mean, DP, Max, Min, MinZ, FO, S, E, H, D)
Descritores1 <- as.data.frame(Descritores1)
Descritores1
Descritores1 <- Descritores1[order(-Descritores1$Mean), ] #ordena do maio para o menor "-"
#Descritores1 <- Descritores1 %>% rownames_to_column(var="Espécies") #da nome a primeira col
SomaTotalD <- apply(Descritores1,2,sum)
SomaTotalD
MediaTotalD <- apply(Descritores1,2,mean)
MediaTotalD
DPTotalD <- apply(Descritores1,2,sd)
DPTotalD
Descritores2 <- cbind(SomaTotalD, MediaTotalD, DPTotalD)
Descritores2 <- as.data.frame(Descritores2)
Descritores2 <- t(Descritores2)
Descritores2
DescritoresFinal <- rbind(Descritores1, Descritores2)
DescritoresFinal
DescritoresFinal <- round (DescritoresFinal, 2)
DescritoresFinal

# Salvando DescritoresFinal
write.table(data.frame("SPP"=rownames(DescritoresFinal),
                       DescritoresFinal),
            "res_ms/DescritoresSPP.txt",
            row.names=FALSE,
            sep="\t")

###TABELA FINAL DESCRITORES----
library(gt)
df <- DescritoresFinal
ncol(df); nrow(df) #no. de N colunas x M linhas
df <- cbind(SPP = rownames(df), df)
gt <- gt(df, rowname_col = "SPP",
   caption = "Descritores da diversidade por espécie (colunas). Sum, soma; RA, abundância rel
gt
gtsave(gt, "res_ms/DescritoresSPP.html")
gtsave(gt, "res_ms/DescritoresSPP.png")

```

```r
###NORMALIDADE----
library(moments)
Assimetria <- apply(m_trab,1,skewness)
Assimetria
Curtose <- apply(m_trab,1,kurtosis)
Curtose
Normalidade1 <- cbind(Assimetria, Curtose)
Normalidade1 <- as.data.frame(Normalidade1)
Normalidade1
SomaTotalN <- apply(Normalidade1,2,sum)
SomaTotalN
MediaTotalN <- apply(Normalidade1,2,mean)
MediaTotalN
DPTotalN <- apply(Normalidade1,2,sd)
DPTotalN
Normalidade2<-cbind(SomaTotalN, MediaTotalN, DPTotalN)
Normalidade2<-as.data.frame(Normalidade2)
Normalidade2 <- t(Normalidade2) #"t" transpoe a matriz
Normalidade2
NormalidadeFinal <- rbind(Normalidade1, Normalidade2)
NormalidadeFinal
NormalidadeFinal <- round(NormalidadeFinal, 2)
NormalidadeFinal

# Salvando NormalidadeFinal
write.table(data.frame("SPP"=rownames(NormalidadeFinal),
                       NormalidadeFinal),
            "res_ms/NormalidadeSPP.txt",
            row.names=FALSE,
            sep="\t")

###TABELA FINAL NORMALIDADE----
nf <- NormalidadeFinal
ncol(nf); nrow(nf) #no. de N colunas x M linhas
nf <- cbind(SPP = rownames(nf), nf)
gt <- gt(nf, rowname_col = "SPP", caption = "Descritores da normalidade por SPP (COLUNA)")
gt
gtsave(gt, "res_ms/NormalidadeSPP.html")

#DIVERSIDADE EM DENSIDADE MATRIZ NÃO-TRANSPOSTA----

m_trab <- (m_dens) #HILL

```

```r
# Números de Hill
Hill <- renyi(m_trab,
              scales = c(0:5),
              hill = TRUE)
Hill
library(ggplot2)
library(tidyr)
library(tidyverse)
grafico1 <- Hill %>%
  rownames_to_column() %>%
  pivot_longer(-rowname) %>%
  mutate(name = factor(name, name[1:length(Hill)])) %>%
  ggplot(aes(x = name, y = value, group = rowname,
             col = rowname)) +
  geom_point(size = 2) +
  geom_line(size = 1) +
  xlab("Parâmetro de ordem de diversidade (q)") +
  ylab("Diversidade") +
  labs(col = "Locais") +
  theme_bw() +
  theme(text = element_text(size = 10)) #ajustar a fonte caso nao caiba no output html
grafico1
ggsave(grafico1, dpi = 300, filename = "res_ms/fig-hill.png")

# Gráfico de abunância
abund <- colSums(m_trab)
#abund <- m_trab[1, ] #escolhe a primeira linha para a distribuição de abundância
df <- data.frame(sp = colnames(m_trab),
                 abun = abund)
grafico2 <- ggplot(df, aes(fct_reorder(sp, -abun),
                           abun, group = 1)) +
  geom_col() +
  geom_line(col = "red", linetype = "dashed") +
  geom_point(col = "red") +
  xlab("Espécies") +
  ylab("Abundância") +
  theme_bw() +
  theme(axis.text.x = element_text(
    angle = 45,
    hjust = 1,
    face = "italic",
    size = 10),
    axis.text.y = element_text(size = 10),
```

```
    axis.title.x = element_text(size = 12),
    axis.title.y = element_text(size = 12))
grafico2
ggsave(grafico2, dpi = 300, filename = "res_ms/fig-abun.png")

# Curvas de rarefação
m_trab_r <- mutate(as.data.frame(m_trab), across(everything(), ceiling)) #arredonda a matriz
colSums(t(m_trab_r))
colnames(t(m_trab_r))
#rarefa <- t(m_trab_r[c("S-R-CT2","S-R-CP2","S-A-TA2","B-A-MU2"),]) #curva de rarefação para
class(m_trab)

# Curva de rarefação para todos os sítios
rarefa <- t(m_trab_r)
## Curva de rarefação para as 8 UA's com maior soma
#m_trab_r <- as.data.frame(t(m_trab_r)) #transpõe a matriz
#col_sums <- colSums(m_trab_r)
#largest_columns <- names(sort(col_sums, decreasing = TRUE)[1:8])
#rarefa <- m_trab_r[largest_columns] #curva de rarefação para as 8 UA's com maior soma
library(iNEXT)
out <- iNEXT(rarefa, q = 0,
             datatype = "abundance",
             size = NULL,
             endpoint = 2000, #define o comprimento de eixo x
             knots = 40,
             se = TRUE,
             conf = 0.95,
             nboot = 50)
grafico3 <- ggiNEXT(out, type = 1, facet.var="None") +
  theme_bw() +
  labs(fill = "Áreas") +
  xlab("Número de indivíduos") +
  ylab("Riqueza de espécies") +
  theme(legend.title=element_blank()) #ver como fica com facet.var="Assemblage"
#grafico3
#ggsave(grafico3, dpi = 300, filename = "fig-rare1.png")

# Curvas de acumulação de espéces
acumula <- specaccum(m_trab,
                     method = "random")
acumula
plot(acumula)
plot_data <- data.frame("UAs" = c(0, acumula$sites),
```

```
                          "Riqueza" = c(0, acumula$richness),
                          "lower" = c(0, acumula$richness - acumula$sd),
                          "upper" = c(0, acumula$richness + acumula$sd))
gLocais <- ggplot(plot_data, aes(x = UAs, y = Riqueza)) +
  geom_point(color = "blue", size = 4) +
  geom_line(color = "blue", lwd = 2) +
  geom_ribbon(aes(ymin = lower, ymax = upper),
              linetype=2, alpha=0.3, fill = "yellow") +
  ylab("Riqueza acumulada") +
  theme_classic() +
  theme(text = element_text(size = 16))
gLocais
ggsave(gLocais, dpi = 300, filename = "res_ms/fig-acum.png")

plot_data <- data.frame("Individuos" = c(0, acumula$individuals),
                          "Riqueza" = c(0, acumula$richness),
                          "lower" = c(0, acumula$richness - acumula$sd),
                          "upper" = c(0, acumula$richness + acumula$sd))
gInd <- ggplot(plot_data, aes(x = Individuos, y = Riqueza)) +
  geom_point(color = "blue", size = 4) +
  geom_line(color = "blue", lwd = 2) +
  geom_ribbon(aes(ymin = lower, ymax = upper),
              linetype=2, alpha=0.3, fill = "yellow") +
  ylab("Riqueza acumulada") +
  theme_classic() +
  theme(text = element_text(size = 16))
gInd

#RIQUEZA RAREFACIONADA----

# Rarefação baseada na matriz de contagem
library(vegan)
data <- m_cont #raremax precisa de numeros inteiros
S <- specnumber(data) #observed number of species
S
#raremax <- min(rowSums(data))
raremax <- round(mean(rowSums(data)))
#raremax <- 1000
raremax
Srare <- rarefy(data, raremax)
Srare
plot(S, Srare, xlab = "Observed No. of Species", ylab = "Rarefied No. of Species")
abline(0, 1)
```

```r

#png("res_ms/fig-rarecurve.png")
rarecurve(data, step = 20,
          sample = raremax, #linhas de rarefação
          col = "blue", cex = 0.6,
          xlab = "Sample size", ylab = "Taxa", xlim = c()) #c(0,5000)
#dev.off()

##TABELA DA COMUNIDADE----

library(dplyr)
library(tidyr)
m_trab <- m_dens %>%
  rename_with(~ gsub("_", ".", .))
m <- m_trab %>%
  group_by(Rio = t_grps$Rio,
           Periodo = t_grps$Periodo,
           Ponto = t_grps$PontoN) %>%
  summarise(across(where(is.numeric),
                   list(mean = mean, sd = sd)),
            .groups = 'drop') %>%
  pivot_longer(
    cols = -c(Rio, Periodo, Ponto),
    names_to = c("Variable", ".value"),
    names_sep = "_"
  )
m
m <- as.data.frame(m)
m_wide <- m %>%
  mutate(stat_string = ifelse(
    Variable == "nome_da_variavel",  #escolhe uma variável pra ter 3 casas decimais
    paste0(round(mean, 3), " (", round(sd, 3), ")"),
    paste0(round(mean, 1), " (", round(sd, 1), ")")
  )) %>%
  unite("Location", Rio, Periodo, Ponto, sep = "_") %>%
  dplyr::select(Variable, Location, stat_string) %>% #dplyr dá conflito com MASS
  pivot_wider(names_from = Location, values_from = stat_string)

m_wide
m_wide <- as.data.frame(m_wide)
m_wide

# Salvado m_wide
```

```r
write.table(m_wide, "res_ms/m_wide_spp.txt",
            sep = ";", dec = ".", #"\t",
            row.names = TRUE,
            quote = TRUE,
            append = FALSE)
m_wide_spp <- read.table("res_ms/m_wide_spp.txt",
                  sep = ";", dec = ".",
                  row.names = 1,
                  header = TRUE,
                  na.strings = NA)

# Exportando dados para Excel
library(openxlsx)
#write.xlsx(m_wide_spp, file = "res_ms/tabela da comunidade.xlsx", rowNames = FALSE)
wb <- loadWorkbook("res_ms/tabela da comunidade.xlsx")
writeData(wb, sheet = "Sheet 1", x = m_wide_spp)
saveWorkbook(wb, "res_ms/tabela da comunidade.xlsx", overwrite = TRUE)

# Tabela final ajustada no Excel
library(readxl)
spp_gttable <- read_excel(
  path  = "res_ms/tabela da comunidade.xlsx",
  sheet = "tabela_final",
  col_names = T,
  range = "A1:AE61") #ajustar para o tamanho da tabela final
spp_gttable
library(gt)
spp_gttable <- gt(spp_gttable)
spp_gttable <- sub_missing(spp_gttable,
                       columns = everything(), missing_text = "")
#spp_gttable <- fmt_number(spp_gttable,
#                          columns = "F.O.(%)", decimals = 1)
spp_gttable
gtsave(spp_gttable, "res_ms/gt-spp_gttable_xlsx.html")
saveRDS(spp_gttable, "res_ms/gt-spp_gttable_xlsx.rds")

#LISTA DE ESPÉCIES

# Exportando dados para Excel
library(openxlsx)
#write.xlsx(m_wide_spp, file = "res_ms/tabela da comunidade.xlsx", rowNames = FALSE)
wb <- loadWorkbook("res_ms/tabela da comunidade.xlsx")
writeData(wb, sheet = "Sheet 1", x = m_wide_spp)
```

```r
saveWorkbook(wb, "res_ms/tabela da comunidade.xlsx", overwrite = TRUE)

# Tabela final ajustada no Excel
library(readxl)
spp_gttable <- read_excel(
  path  = "dados/rebio23-zoopl.xlsx",
  sheet = "Espécies",
  col_names = T,
  range = "L1:S90") #ajustar para o tamanho da tabela final
spp_gttable
library(gt)
spp_gttable <- gt(spp_gttable)
spp_gttable <- sub_missing(spp_gttable,
                       columns = everything(), missing_text = "")
#spp_gttable <- fmt_number(spp_gttable,
#                          columns = "F.O.(%)", decimals = 1)
spp_gttable
gtsave(spp_gttable, "res_ms/gt-checklist_gttable_xlsx.html")
saveRDS(spp_gttable, "res_ms/gt-checklist_gttable_xlsx.rds")
```

```r
# Escolher sumário de uma variavel
m
var <- "h.roots"
m[m$Variable == var, "mean"] #cada valor de var
summary(m[m$Variable == var, "mean"]) #sumário dos valores de var

# Escolher sumário de um grupo de variáveis do df m
vars <- unique(grep("^h\\.", m$Variable, value = TRUE))
summaries <- list() #criam uma lista vazia para guardar os sumários
# Loop para cada variável do grupo e guarda em summaries
for (var in vars) {
  summaries[[var]] <- summary(m[m$Variable == var, "mean"])
}
#var is a temporary variable used in the for loop to iterate through
#each variable name that starts with "h."

summaries
summary_table <- do.call(rbind, lapply(summaries, as.data.frame.list))
round(summary_table, 2)
#sink(file = "summary_h.txt", split = TRUE)
round(summary_table[order(summary_table$Mean, decreasing = FALSE), ], 2)
#sink()
```

```r
# Tabela limpa
#summary_table <- cbind(Variable = rownames(summary_table), summary_table)
#rownames(summary_table) <- NULL
#colnames(summary_table) <- c("Variable", "Min", "Q1", "Median", "Mean", "Q3", "Max")
#summary_table
```

Rarefaction analysis showed that the Caiana stream had a taxa richness slightly higher than that of the Barro Branco stream (Figure 4). Nevertheless, given a similar sample size, the Barro Branco stream would have a greater diversity, as shown by the Chao index that estimated an extrapolated richness of 58 (±8.9) species for the Barro Branco stream, compared with 54.2 (± 4.3) species for the Caiana stream.

```r
#ZOOREBIO_part - CURVA DE ACUMULAÇÃO----
#####....----

##ORGANIZANDO DADOS----

dev.off() #apaga os graficos, se houver algum
rm(list=ls(all=TRUE)) #limpa a memória
cat("\014") #limpa o console

m_cont <- read.csv("res_ms/m_cont.csv",
                   sep = ";", dec = ".",
                   row.names = 1,
                   header = TRUE,
                   na.strings = NA)
m_trab <- m_cont

library(tidyverse)

m_trab_r <- mutate(m_trab, across(everything(), ceiling)) #arredonda a matriz original para

colSums(t(m_trab_r))
colnames(t(m_trab_r))

ca_abund <- colSums(m_trab_r[grep("-C-", rownames(m_trab_r)), ])
bb_abund  <- colSums(m_trab_r[grep("-B-", rownames(m_trab_r)), ])

m_rios <- rbind("Caiana" = ca_abund, "Barro Branco" = bb_abund)
m_rios

rarefa <- t(m_rios)

```

```r
library(iNEXT)
out <- iNEXT(rarefa, q = 0,
             datatype = "abundance",
             size = NULL,
             endpoint = 15000, #define o comprimento de eixo x
             knots = 40,
             se = TRUE,
             conf = 0.95,
             nboot = 50)

library(ggplot2)
grafico3 <- ggiNEXT(out, type = 1, facet.var="None") + #ver como fica com facet.var="Assembl
  theme_bw() +
  labs(fill = "Áreas") +
  xlab("Number of individuals") +
  ylab("Taxa richness") +
  scale_linetype_discrete(
    labels = c("Interpolation", "Extrapolation")) +
  theme(legend.title = element_blank(),
        legend.position = "bottom",
        legend.text = element_text(size = 14),
        axis.title = element_text(size = 14),
        axis.text = element_text(size = 12)
)
grafico3
ggsave(grafico3, width = 8, height = 6, dpi = 300, filename = "res_ms/fig-rare1.png")

#ESTIMADORES DA DIVERSIDADE

#https://rpubs.com/Juan-Mt/diversity-estimations
#https://analises-ecologicas.com/cap11

library(vegan)
m_cont <- as.matrix(m_cont)
storage.mode(m_cont) <- "numeric"

BB <- m_cont[grepl("^[SC]-B-", rownames(m_cont)),, drop = FALSE]
CA <- m_cont[grepl("^[SC]-C-", rownames(m_cont)),, drop = FALSE]

nrow(BB)
nrow(CA)

# Calculando a riqueza observada
```

```r
BB_total <- colSums(BB)
CA_total <- colSums(CA)
S_BB <- sum(BB_total > 0)
S_CA <- sum(CA_total > 0)
S_BB
S_CA

# Chao1
chao_BB <- estimateR(BB_total)
chao_CA <- estimateR(CA_total)
chao_BB
chao_CA

# Separando Chao1
resultado_chao <- data.frame(
  Stream = c("Barro Branco", "Caiana"),
  Observed = c(
    chao_BB["S.obs"],
    chao_CA["S.obs"]
  ),
  Chao1 = c(
    chao_BB["S.chao1"],
    chao_CA["S.chao1"]
  ),
  SE = c(
    chao_BB["se.chao1"],
    chao_CA["se.chao1"]
  )
)
resultado_chao

# 95% confidence intervals
resultado_chao$Lower95 <- resultado_chao$Chao1 -
  1.96 * resultado_chao$SE
resultado_chao$Upper95 <- resultado_chao$Chao1 +
  1.96 * resultado_chao$SE

# Do not allow the lower limit to be below observed richness
resultado_chao$Lower95 <- pmax(
  resultado_chao$Lower95,
  resultado_chao$Observed
)

```

```r
sink(file = "res_ms/chao.txt")
# Tabela pronta
resultado_chao
sink()
```

Regarding the community composition as shown by the UPGMA (Figure 5), sites did not aggregate to represent the different streams or their hierarchy across seasons. Even though the Cophenetic coefficients showed that the UPGMA dendrograms were a good represenation of the original Bray-Curtis dissimilarities among sampling sites ($CC_{dry} = 0.84$ and $CC_{wet} = 0.87$), the UPGMA cluster analyses revealed low-to-moderate spatial clustering strength for the zooplankton communities across both periods, as indicated by the Agglomerative coefficients ($AC_{dry} = 0.34$ and $AC_{wet} = 0.39$). The PERMANOVA based on Bray–Curtis dissimilarities showed a significant effect of both hydrological period ($F_{1,86} = 5.3, R^2 = 0.056, p = 0.001$) and stream ($F_{1,86} = 1.8, R^2 = 0.019, p = 0.042$) on community composition. The full model including the Season × Stream interaction was also significant ($F_{1,86} = 2.1, R^2 = 0.022, p = 0.031$), explaining 9.73% of the total variation, with 90.27% of the variation unexplained by the factors included in the model.

```r
#ZOOREBIO_part - UPGMA----
#####....----

##ORGANIZANDO DADOS----

dev.off() #apaga os graficos, se houver algum
rm(list=ls(all=TRUE)) #limpa a memória
cat("\014") #limpa o console

t_grps <- read.csv("res_ms/t_grps.csv",
                   sep = ";", dec = ".",
                   row.names = 1,
                   header = TRUE,
                   na.strings = NA)
m_dens <- read.csv("res_ms/m_dens.csv",
                   sep = ";", dec = ".",
                   row.names = 1,
                   header = TRUE,
                   na.strings = NA)
m_hab_part <- read.csv("res_ms/m_hab_part.csv",
                       sep = ";", dec = ".",
                       row.names = 1,
                       header = TRUE,
                       na.strings = NA)
m_cont <- read.csv("res_ms/m_cont.csv",
```

```
                    sep = ";", dec = ".",
                    row.names = 1,
                    header = TRUE,
                    na.strings = NA)

#TRANSFORMANDO A MATRIZ DE DENSIDADE

library(vegan)
m_dens_trns <- asin(sqrt(decostand(m_dens,
                                   method="total", MARGIN = 2)))

write.table(m_dens_trns, "res_ms/m_dens_trns.csv",
            sep = ";", dec = ".", #"\t",
            row.names = TRUE,
            quote = TRUE,
            append = FALSE)
m_dens_trns <- read.csv("res_ms/m_dens_trns.csv",
                    sep = ";", dec = ".",
                    row.names = 1,
                    header = TRUE,
                    na.strings = NA)

vegdist <- vegdist(m_dens_trns, method = "bray",
                    diag = TRUE,
                    upper = FALSE)
cluster <- hclust(vegdist, method = "average")
plot (cluster, main = "Cluster Dendrogram - Bray-Curtis")
rect.hclust(cluster, k = 3, h = NULL)
#h = 0.8 fornece os grupos formados na altura h
as.matrix(vegdist)[1:6, 1:6]

#SEPARANDO OS PERÍODOS E REFAZENDO A TRANSFORMAÇÃO

# Fazendo a média por ponto
ponto_id <- sub("-H[0-9]+$", "", rownames(m_dens))
m_dens_media <- aggregate(m_dens, by = list(Ponto = ponto_id), FUN = mean)
rownames(m_dens_media) <- m_dens_media$Ponto
m_dens_media$Ponto <- NULL

write.table(m_dens_media, "res_ms/m_dens_media.csv",
            sep = ";", dec = ".", #"\t",
            row.names = TRUE,
            quote = TRUE,
```

```
69              append = FALSE)
70  m_dens_media <- read.csv("res_ms/m_dens_media.csv",
71                       sep = ";", dec = ".",
72                       row.names = 1,
73                       header = TRUE,
74                       na.strings = NA)
75  m_dens_media <- m_dens_trns
76  # Separar por período
77  seco_med <- m_dens_media[startsWith(rownames(m_dens_media), "S-"), ]
78  chuv_med <- m_dens_media[startsWith(rownames(m_dens_media), "C-"), ]
79  # Filtrar COLUNAS zeradas (espécies ausentes no período)
80  seco_med <- seco_med[, colSums(seco_med) > 0]
81  chuv_med <- chuv_med[, colSums(chuv_med) > 0]
82  # Filtrar LINHAS zeradas (pontos sem organismos)
83  seco_med <- seco_med[rowSums(seco_med) > 0, ]
84  chuv_med <- chuv_med[rowSums(chuv_med) > 0, ]
85  # Transformação
86  seco_trns <- asin(sqrt(decostand(seco_med,
87                                   method="total", MARGIN = 2)))
88  chuv_trns <- asin(sqrt(decostand(chuv_med,
89                                   method="total", MARGIN = 2)))
90  # Distâncias de Bray-Curtis e UPGMA
91  dist_seco <- vegdist(seco_trns, method = "bray")
92  dist_chuv <- vegdist(chuv_trns, method = "bray")
93  upgma_seco <- hclust(dist_seco, method = "average")
94  upgma_chuv <- hclust(dist_chuv, method = "average")
95
96  # Coeficientes de aglomeração
97  agg_coef_seco <- coef.hclust(upgma_seco)
98  agg_coef_chuv <- coef.hclust(upgma_chuv)
99  # Gerar a matriz de distância cofenética a partir da árvore UPGMA
100 cofe_seco <- cophenetic(upgma_seco)
101 cofe_chuv <- cophenetic(upgma_chuv)
102 # Calcular a correlação com a matriz de distância original (Bray-Curtis)
103 cor_seco <- cor(dist_seco, cofe_seco)
104 cor_chuv <- cor(dist_chuv, cofe_chuv)
105 # Visualizar os resultados
106 cat("Correlação Cofenética (Seco):", round(cor_seco, 3), "\n")
107 cat("Correlação Cofenética (Chuvoso):", round(cor_chuv, 3), "\n")
108
109 # Plot dos dendrogramas agrupados por ponto
110 upgma_seco$labels <- sub("^[A-Z]-", "", upgma_seco$labels)
111 upgma_chuv$labels <- sub("^[A-Z]-", "", upgma_chuv$labels)
```

```r
dend_seco <- as.dendrogram(hclust(dist_seco, method = "average"))
dend_chuv <- as.dendrogram(hclust(dist_chuv, method = "average"))

#png("res_ms/fig-upgma.png", width=11, height=9, units="in", res=400)
par(mfrow = c(1, 2), mar = c(5, 4, 2, 6)) #B, L, T, R
plot(as.dendrogram(upgma_seco),
     horiz = TRUE,
     xlab = "Bray-Curtis dissimilariry",
     xlim = c(max(attr(dend_seco, "height")), 0),
     cex.lab = 1)
title(main = "Dry", line = 0, cex.main = 1.2)
mtext(paste0("Agglomerative Coefficient = ", round(agg_coef_seco, 2)),
      side = 2,           # Lado esquerdo (eixo Y)
      line = 1,           # Distância da margem
      at = 7,             # Posição na base do eixo
      las = 0,            # Orientação do texto
      cex = 0.8)          # Tamanho da fonte
mtext(paste0("Cophenetic Coefficient = ", round(cor_seco, 2)),
            side = 2,           # Lado esquerdo (eixo Y)
            line = 2,           # Distância da margem
            at = 7,             # Posição na base do eixo
            las = 0,            # Orientação do texto
            cex = 0.8)          # Tamanho da fonte

plot(as.dendrogram(upgma_chuv),
     horiz = TRUE,
     xlab = "Bray-Curtis dissimilariry",
     xlim = c(max(attr(dend_chuv, "height")), 0),
     cex.lab = 1)
title(main = "Wet", line = 0, cex.main = 1.2)
mtext(paste0("Agglomerative Coefficient = ", round(agg_coef_chuv, 2)),
      side = 2,           # Lado esquerdo (eixo Y)
      line = 1,           # Distância da margem
      at = 7,             # Posição na base do eixo
      las = 0,            # Orientação do texto
      cex = 0.8)          # Tamanho da fonte
mtext(paste0("Cophenetic Coefficient = ", round(cor_chuv, 2)),
      side = 2,           # Lado esquerdo (eixo Y)
      line = 2,           # Distância da margem
      at = 7,             # Posição na base do eixo
      las = 0,            # Orientação do texto
      cex = 0.8)          # Tamanho da fonte
par(mfrow = c(1, 1))
```

```r
#dev.off()

#OUTRA FORMA
#plot(as.dendrogram(upgma_chuv),
#     horiz = TRUE,
#     main = "Wet",
#     xlab = "Bray-Curtis dissimilarity",
#     xlim = c(max(attr(dend_chuv, "height")), 0))
# Adiciona o texto no canto inferior esquerdo
#text(x = max(attr(dend_chuv, "height")) * 0.9,
#     y = 3,
#     labels = paste0("Agglomerative Coefficient = ", round(agg_coef_chuv, 2)),
#     adj = c(0, 0),
#     cex = 0.85,
#     srt = 90) # srt = 90 gira o texto para ficar vertical
#OU
#mtext(paste0("Cophenetic Coefficient = ", round(cor_seco, 2)),
#      side = 2,    # 1 = bottom margin
#      line = 2.5,    # Line distance below axis (increase if it overlaps numbers)
#      adj = 0.36,     # Posição horizontal
#      cex = 0.85)  # Font size
```

```r
#ORGANIZANDO DADOS----

dev.off()
rm(list=ls(all=TRUE))
cat("\014")

t_grps <- read.csv("res_ms/t_grps.csv",
                   sep = ";", dec = ".",
                   row.names = 1,
                   header = TRUE,
                   na.strings = NA)
m_dens_trns <- read.csv("res_ms/m_dens_trns.csv",
                        sep = ";", dec = ".",
                        row.names = 1,
                        header = TRUE,
                        na.strings = NA)

colnames(t_grps)
m_trab <- m_dens_trns
library(vegan)

```

```r
#MRPP----

#TRABALHANDO COM MULTIPLAS VARIAVEIS----

# Run MRPP for multiple variables
fatores <- c("Periodo", "Rio", "Ref_site")
set.seed(123)
mrpp_results <- lapply(fatores, function(v) {
  grp <- as.factor(t_grps[[v]])
  mrpp(
    dat = m_trab,
    grouping = grp,
    distance = "bray"
  )
})
names(mrpp_results) <- fatores
mrpp_results
# Print one result
mrpp_results$Ref_site
# Loop through summaries
sink(file = "res_ms/mrpp-dens.txt", split = TRUE)#, append = TRUE)
for (v in names(mrpp_results)) {
  cat("\n============================\n")
  cat("MRPP for:", v, "\n")
  print(mrpp_results[[v]])
}
sink()
# Extract key statistics into a table
mrpp_summary <- data.frame(
  Variable = fatores,
  Delta = sapply(mrpp_results, function(x) x$delta),
  Expected_Delta = sapply(mrpp_results, function(x) x$E.delta),
  A = sapply(mrpp_results, function(x) x$A),
  P_value = sapply(mrpp_results, function(x) x$Pvalue)
)
sink("res_ms/mrpp-dens.txt", append = TRUE)
mrpp_summary
sink()

##TRABALHANDO COM UMA UNICA VARIAVEL----

# Run MRPP for a single variable
grp <- as.factor(t_grps$Ref_site) #UA, mes, estacao, area, ambiente
```

```r
set.seed(123)
mrpp <- mrpp(dat = m_trab, grouping = grp, distance = "bray")
mrpp
summary(mrpp)

# Teste de hipótese para uma unica variavel
str(mrpp)
p_value <- mrpp$Pvalue
print(sprintf("p-value: %.10f", p_value))
# Conditional statement to check the p-value and print the appropriate message
if (p_value < 0.05) {
  print(sprintf("Existe diferença significativa porquê o valor de p é %.10f, sendo menor que
} else {
  print(sprintf("Não existe diferença significativa porquê o valor de p é %.10f, sendo maior
}

##TESTES PAREADOS----

# Todos os pares de níveis do fator
pairs <- combn(levels(as.factor(grp)), 2, simplify = FALSE)
pairs
# Aplicar mrpp para cada par
res_list <- lapply(pairs, function(p) {
  sel <- grp %in% p
  mrpp(dat = m_trab[sel, ], grouping = droplevels(grp[sel]), distance = "bray")
})
# Dar nomes aos resultados
names(res_list) <- sapply(pairs, function(p) paste(p, collapse = "_vs_"))
# visualizar
res_list
# Resumo
summary_tab <- data.frame(
  Comparison = names(res_list),
  A = sapply(res_list, function(x) x$A),
  Delta = sapply(res_list, function(x) x$delta),
  Pvalue = sapply(res_list, function(x) x$Pvalue)
)
sink("res_ms/mrpp-dens.txt", append = TRUE)
summary_tab
sink()

###MATRIX-STYLE TABLE----
library(dplyr)
```

```r
library(tidyr)
library(knitr)
library(kableExtra)
# Extract unique group names from the comparisons
groups1 <- sort(unique(unlist(strsplit(summary_tab$Comparison, "_vs_"))))
# Initialize empty matrix
pval_matrix <- matrix(NA, nrow = length(groups1), ncol = length(groups1),
                      dimnames = list(groups1, groups1))
# Fill matrix with Pvalues
for (i in 1:nrow(summary_tab)) {
  comp <- unlist(strsplit(summary_tab$Comparison[i], "_vs_"))
  g1 <- comp[1]
  g2 <- comp[2]
  p <- summary_tab$Pvalue[i]

  pval_matrix[g1, g2] <- p
  pval_matrix[g2, g1] <- p  #make symmetric
}

# Format p-values: red if <0.05
pval_colored <- ifelse(is.na(pval_matrix), "",
                       ifelse(pval_matrix < 0.05,
                              paste0("<span style='color:red;'>", sprintf("%.3f", pval_matri
                              sprintf("%.3f", pval_matrix)))

# Convert to data.frame for kable
pval_df <- as.data.frame(pval_colored)
pval_df <- tibble::rownames_to_column(pval_df, "Group")
# Render HTML table
matrixstyle <- pval_df %>%
  kable("html", escape = FALSE) %>%
  kable_styling(full_width = FALSE)
matrixstyle
save_kable(matrixstyle, "res_ms/kb-mrpp-dens_matrixstyle.html")

##GRÁFICO DA MRPP----

png("res_ms/fig-mrpp-dens.png", width = 15, height = 10, units = "in", res = 300)
def.par <- par(no.readonly = TRUE)
layout(matrix(1:2,nr=1))
plot(ord <- metaMDS(m_trab, distance="bray"), type="text", display="sites")
with(t_grps, ordihull(ord, grp))
with(mrpp, {
```

```r
  fig.dist <- hist(boot.deltas, xlim=range(c(delta,boot.deltas)),
                   main="Test of Differences Among Groups")
  abline(v=delta);
  text(delta, 2*mean(fig.dist$counts), adj = -0.5,
       expression(bold(delta)), cex=1.5 )  }
)
par(def.par)
dev.off()

###AVALIANDO AS DISTÂNCIAS MÉDIAS----
md <- with(t_grps, meandist(vegdist(m_trab), grp))
md
summary(md)
par(mfrow=c(1,2))
plot(md)
plot(md, kind="histogram")
par(mfrow=c(1,1))

#PerMANOVA SIMPLIFICADA----

#browseURL("https://riffomonas.org/code_club/2021-03-18-vegan")
#browseURL("https://riffomonas.org/code_club/2021-03-22-reproducibility")

library(vegan)

## TESTES PARA VARIAVEIS COM MAIS DE DOIS FATORES

adonis2(m_trab ~ Ref_site, data = t_grps, method = "bray")
set.seed(123)
perm <- 999
all_test <- adonis2(m_trab ~ Ref_site, data = t_grps, method = "bray", permutations = perm)
all_test

## TESTES PARA VARIAVEIS COM DOIS FATORES

sink(file = "res_ms/perma-dens.txt", split = TRUE)#, append = TRUE)
set.seed(123)
perm <- 999
adonis2(m_trab ~ Periodo, data = t_grps, method = "bray", permutations = perm)
adonis2(m_trab ~ Rio, data = t_grps, method = "bray", permutations = perm)
adonis2(m_trab ~ Periodo + Rio, data = t_grps, #+ no lugar de * tira os residuos
        method = "bray", permutations = perm)

```

```r
## AVALIANDO INTERAÇÕES-----

adonis2(m_trab ~ Periodo*Rio, data = t_grps, method = "bray", permutations = perm,
        by = "terms") #ou by=NULL remove as interações
adonis2(m_trab ~ Periodo*Rio, data = t_grps, method = "bray", permutations = perm,
        by = "margin") #ou by=NULL remove as interações
adonis2(m_trab ~ Periodo*Rio, data = t_grps, method = "bray", permutations = perm,
        by = NULL) #ou by=NULL remove as interações
sink()

#BETADISPER

dist_bray <- vegdist(m_trab, method = "bray")
t_grps$Periodo_Rio <- interaction(
  t_grps$Periodo,
  t_grps$Rio,
  sep = "_")
bd <- betadisper(dist_bray, t_grps$Periodo_Rio)
permutest(bd, permutations = 999)

# PARA CADA FATOR

bd_periodo <- betadisper(dist_bray, t_grps$Periodo)
bd_rio <- betadisper(dist_bray, t_grps$Rio)
permutest(bd_periodo, permutations = 999)
permutest(bd_rio, permutations = 999)
```

Hierarchical additive partitioning was firstly performed separate for ech season with the gamma diversity including both study stream. The comparison between dry and wet seasons revealed that zooplankton diversity was structured non-randomly across spatial scales ($p < 0.001$ for all components except β_2; Table 4). For the dry season, the alpha diversity ($\alpha_1 = 12.5\%$) was significantly lower than expected by chance (SES $= -22.1, p = 0.001$), indicating spatial aggregation or environmental filtering at finer scales. Habitat-scale beta diversity ($\beta_1 = 9.7\%$) was significantly lower than expected under the null model (SES $= -8.9, p = 0.001$), reflecting high compositional similarity among local sampling units. On the other hand, beta diversity, among streams ($\beta_3 = 26.7\%$) was significantly higher than that predicted by chance (SES $= 5.6, p = 0.001$), demonstrating substantial species turnover at the highest hierarchical level. The intermediate scale (among sites) contributed the most for the observed regional diversity ($\beta_2 = 51.1\%$) and it did not deviate significantly from null expectations (SES $= 0.6, p = 0.519$). Contrasting results were obtained for the wet season, when the observed alpha diversity ($\alpha_1 = 23.7\%$) was much higher than the dry season, yet significantly lower than that expected by chance (SES $= -45.4, p = 0.001$). Similar to results from the dry season, the habitat-scale beta diversity ($\beta_1 = 11.8\%$) was significantly lower than expected under the null

model (SES $= -7.0, p = 0.001$), and the the beta diversity among streams ($\beta_3 = 16.3\%$) was significantly higher than that predicted by chance (SES $= 9.5, p = 0.001$). The intermediate scale (among sites) gave a lesser contribution to the total observed diversity ($\beta_2 = 48.3\%$), when compared to the dry season and, differently from the dry season, it deviated significantly from the null expectation (SES $= 13.4, p = 0.001$).

Hierarchical additive partitioning was performed separately for each season within the Barro Branco and Caiana streams (Figure 7). Across all four combinations, local alpha diversity ($\alpha_1 = 14.7\% - 30.8\%$) and fine-scale habitat beta diversity ($\beta_1 = 12.1\% - 14.8\%$) were consistently lower than expected under null model predictions (see Table 4 for SES and P-values). The alpha diversity was higher during the wet season when compared with the dry season across both streams. This recurring negative deviation highlights strong local spatial aggregation and compositional homogenization among sampling points within sites. Beta diversity among sites ($\beta_2 = 54.6\% - 73.2\%$) consistently contributed the largest share of total regional richness and was significantly higher than predicted by chance (see Table 4 for SES and P-values). These positive effect sizes confirm that spatial species turnover among sites is the primary driver structuring zooplankton assemblages within both streams regardless of seasonality.

```
#ZOOREBIO_part - ADDPART----
#####....----

##ORGANIZANDO DADOS----

dev.off() #apaga os graficos, se houver algum
rm(list=ls(all=TRUE)) #limpa a memória
cat("\014") #limpa o console

m_esc <- read.csv("res_ms/m_esc.csv",
                  sep = ";", dec = ".",
                  row.names = 1,
                  header = TRUE,
                  na.strings = NA)
m_dens <- read.csv("res_ms/m_dens.csv",
                  sep = ";", dec = ".",
                  row.names = 1,
                  header = TRUE,
                  na.strings = NA)
m_cont <- read.csv("res_ms/m_cont.csv",
                  sep = ";", dec = ".",
                  row.names = 1,
                  header = TRUE,
                  na.strings = NA)
```

```r

#BUSCANDO AS ESCALAS

# Partição total
colnames(m_esc)
pt.total <- m_esc[, c("pt.rio", "pt.periodo", "pt.rebio")]
head(pt.total)
dim(pt.total)

# Partição seco
pt.seco <- m_esc[, c("s.habitat", "s.ponto", "s.rio", "s.rebio")]
pt.seco <- pt.seco[grep("^S-", rownames(pt.seco)), ]
head(pt.seco)
dim(pt.seco)
m_cont_s <- m_cont[grep("^S-", rownames(m_cont)), ]

# Partição chuvoso
pt.chuva <- m_esc[, c("c.habitat", "c.ponto", "c.rio", "c.rebio")]
pt.chuva <- pt.chuva[grep("^C-", rownames(pt.chuva)), ]
head(pt.chuva)
dim(pt.chuva)
m_cont_c <- m_cont[grep("^C-", rownames(m_cont)), ]

# Partição bb.seco (A)
bb.seco <- m_esc[, c("bb.habitat", "bb.ponto", "bb.rio")]
bb.seco <- bb.seco[grep("S-B-", rownames(bb.seco)), ]
head(bb.seco)
dim(bb.seco)
m_cont_bb.seco <- m_cont[grep("S-B-", rownames(m_cont)), ]

# Partição bb.chuva (B)
bb.chuva <- m_esc[, c("bb.habitat", "bb.ponto", "bb.rio")]
bb.chuva <- bb.chuva[grep("C-B-", rownames(bb.chuva)), ]
head(bb.chuva)
dim(bb.chuva)
m_cont_bb.chuva <- m_cont[grep("C-B-", rownames(m_cont)), ]

# Partição ca.seco (C)
ca.seco <- m_esc[, c("ca.habitat", "ca.ponto", "ca.rio")]
ca.seco <- ca.seco[grep("S-C-", rownames(ca.seco)), ]
head(ca.seco)
dim(ca.seco)
m_cont_ca.seco <- m_cont[grep("S-C-", rownames(m_cont)), ]
```

```r

# Partição ca.chuva (D)
ca.chuva <- m_esc[, c("ca.habitat", "ca.ponto", "ca.rio")]
ca.chuva <- ca.chuva[grep("C-C-", rownames(ca.chuva)), ]
head(ca.chuva)
dim(ca.chuva)
m_cont_ca.chuva <- m_cont[grep("C-C-", rownames(m_cont)), ]

#PARTIÇÃO

#pt.total (m_cont)
#pt.seco (m_cont_s), pt.chuva (m_cont_c)
#bb.seco(A) (m_cont_bb.seco), bb.chuva(B) (m_cont_bb.chuva)
#ca.seco(C) (m_cont_ca.seco), ca.chuva(D) (m_cont_ca.chuva)

library(vegan)
part1 <- ca.chuva
cont <- m_cont_ca.chuva
sink(file = "res_ms/part-pt.ca.chuva.txt")
#VALORES ABSOLUTOS
set.seed(666)
part.a <- adipart(y = cont, x = part1, index = "richness", relative = FALSE,
                  weights = "unif", nsimul = 999)
part.a

#EM PORCENTAGEM
set.seed(666)
part.p <- adipart(y = cont, x = part1, index = "richness", relative = TRUE,
                  weights = "unif", nsimul = 999)
part.p

#saveRDS(part.p, "res_ms/part.p-pt.total.rds")
#saveRDS(part.p, "res_ms/part.p-pt.seco.rds")
#saveRDS(part.p, "res_ms/part.p-pt.chuva.rds")
#saveRDS(part.p, "res_ms/part.p-bb.seco.rds")
#saveRDS(part.p, "res_ms/part.p-bb.chuva.rds")
#saveRDS(part.p, "res_ms/part.p-ca.seco.rds")
#saveRDS(part.p, "res_ms/part.p-ca.chuva.rds")

#part.p <- readRDS("res_ms/part.p-pt.total.rds")
#part.p <- readRDS("res_ms/part.p-pt.seco.rds")
#part.p <- readRDS("res_ms/part.p-pt.chuva.rds")
#part.p <- readRDS("res_ms/part.p-bb.seco.rds")
```

```r
#part.p <- readRDS("res_ms/part.p-bb.chuva.rds")
#part.p <- readRDS("res_ms/part.p-ca.seco.rds")
#part.p <- readRDS("res_ms/part.p-ca.chuva.rds")

# Extrair valores do objeto part.p
part2 <- part.p

statistic <- part2$statistic
SES <- part2$oecosimu$z
mean <- part2$oecosimu$means
lower_2.5 <- apply(part2$oecosimu$simulated, 1, quantile, probs = 0.025)
median_50 <- apply(part2$oecosimu$simulated, 1, quantile, probs = 0.50)
upper_97.5 <- apply(part2$oecosimu$simulated, 1, quantile, probs = 0.975)
Pr_sim <- part2$oecosimu$pval

# Definir função para adicionar significância
get_significance <- function(p) {
  if (p <= 0.001) {
    return("***")
  } else if (p <= 0.01) {
    return("**")
  } else if (p <= 0.05) {
    return("*")
  } else if (p <= 0.1) {
    return(".")
  } else {
    return(" ")
  }
}
significance <- sapply(Pr_sim, get_significance)

# Criar dataframe com todas as informações
diversity_data <- data.frame(
  Component = names(statistic),
  Observed = round(statistic, 2),
  SES = round(SES, 2),
  Mean = round(mean, 2),
  Lower_2.5 = round(lower_2.5, 2),
  Median_50 = round(median_50, 2),
  Upper_97.5 = round(upper_97.5, 2),
  Pr_sim = Pr_sim,
  Sign = significance
)
```

```r

# Exibir a tabela
library(gt)
print(diversity_data)
#str(part2)
gt(diversity_data)

# Criar dataframe com valores observados e esperados
observed_values <- part2$statistic
observed_values
expected_values <- part2$oecosimu$means
expected_values

# Combine observed and expected values into a dataframe
df <- data.frame(
  Component = rep(names(observed_values), 2), # Use names from observed_values
  Type = rep(c("Observed", "Expected"), each = length(observed_values)),
  Value = c(observed_values, expected_values) # Observado primeiro, Esperado depois
)

df$Type <- factor(df$Type, levels = c("Observed", "Expected"))
df
df_filtered <- df[!df$Component %in% c("alpha.2", "alpha.3", "alpha.4", "gamma"), ]
df_filtered$Component <- factor(df_filtered$Component, levels = c("beta.4", "beta.3", "beta.
df_filtered

diversity_data <- df_filtered
diversity_data

# Calcular porcentagens para cada componente
diversity_data <- diversity_data %>%
  group_by(Type) %>%
  mutate(Percentage = Value / sum(Value) * 100)

hier <- hiersimu(y = cont, x = part1, FUN = diversity)
hier
sink()

#saveRDS(diversity_data, "res_ms/part-pt.gtotal.rds")
#saveRDS(diversity_data, "res_ms/part-pt.gseco.rds")
#saveRDS(diversity_data, "res_ms/part-pt.gchuva.rds")
#saveRDS(diversity_data, "res_ms/part-bb.gseco.rds")
#saveRDS(diversity_data, "res_ms/part-bb.gchuva.rds")
```

```r
#saveRDS(diversity_data, "res_ms/part-ca.gseco.rds")
#saveRDS(diversity_data, "res_ms/part-ca.gchuva.rds")

#GRÁFICOS

library(ggplot2)
library(dplyr)

# PARTIÇÃO TOTAL

pt.gtotal <- readRDS("res_ms/part-pt.gtotal.rds")
ggplot_pt.total <- ggplot(pt.gtotal, aes(fill = Component, y = Percentage, x = Type)) +
  geom_bar(position = "stack", stat = "identity") +
  labs(title = "Total",
       y = "% of the total diversity",
       x = "") +
  scale_fill_manual(values = c("alpha.1" = "black",
                               "beta.1" = "darkgray",
                               "beta.2" = "gray",
                               "beta.3" = "lightgray",
                               "beta.4" = gray(0.9)),
                    labels = c("alpha.1" = "Habitat ( 1)",
                               "beta.1"  = "Habitat ( 1)",
                               "beta.2"  = "Site ( 2)",
                               "beta.3"  = "Stream ( 3)",
                               "beta.4"  = " 4")) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 0, hjust = 0.5),
    panel.grid = element_blank(),
    legend.position = "right"
  )
ggplot_pt.total
ggsave(ggplot_pt.total, dpi = 300, filename = "res_ms/fig-part_total.png")

# PARTIÇÃO POR PERÍODO

# Partição seco #pt.seco (m_cont_s)
pt.gseco <- readRDS("res_ms/part-pt.gseco.rds")
ggplot_pt.seco <- ggplot(pt.gseco, aes(fill = Component, y = Percentage, x = Type)) +
  geom_bar(position = "stack", stat = "identity") +
  labs(title = "Dry",
       y = "% of the total diversity",
```

```
        x = "") +
  scale_fill_manual(values = c("alpha.1" = "black",
                               "beta.1" = "darkgray",
                               "beta.2" = "gray",
                               "beta.3" = "lightgray",
                               "beta.4" = gray(0.9)),
                    labels = c("alpha.1" = "Habitat ( 1)",
                               "beta.1"  = "Habitat ( 1)",
                               "beta.2"  = "Site ( 2)",
                               "beta.3"  = "Stream ( 3)",
                               "beta.4"  = " 4")) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5), #hjust 0=left, 0.5=center, 1=right
    axis.text.x = element_text(angle = 0, hjust = 0.5),
    panel.grid = element_blank(),
  )
ggplot_pt.seco
ggsave(ggplot_pt.total, dpi = 300, filename = "res_ms/fig-part_seco.png")

# Partição chuvoso #pt.chuva (m_cont_c)
pt.gchuva <- readRDS("res_ms/part-pt.gchuva.rds")
ggplot_pt.chuva <- ggplot(pt.gchuva, aes(fill = Component, y = Percentage, x = Type)) +
  geom_bar(position = "stack", stat = "identity") +
  labs(title = "Wet",
       y = "% of the total diversity",
       x = "") +
  scale_fill_manual(values = c("alpha.1" = "black",
                               "beta.1" = "darkgray",
                               "beta.2" = "gray",
                               "beta.3" = "lightgray",
                               "beta.4" = gray(0.9)),
                    labels = c("alpha.1" = "Habitat ( 1)",
                               "beta.1"  = "Habitat ( 1)",
                               "beta.2"  = "Site ( 2)",
                               "beta.3"  = "Stream ( 3)",
                               "beta.4"  = " 4")) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5), #hjust 0=left, 0.5=center, 1=right
    axis.text.x = element_text(angle = 0, hjust = 0.5),
    panel.grid = element_blank(),
  )
```

```r
ggplot_pt.chuva
ggsave(ggplot_pt.total, dpi = 300, filename = "res_ms/fig-part_chuva.png")

# PARTIÇÃO POR RIO

# bb.seco (A) #bb.seco(A) (m_cont_bb.seco)
bb.gseco <- readRDS("res_ms/part-bb.gseco.rds")
ggplot_bb.seco <- ggplot(bb.gseco, aes(fill = Component, y = Percentage, x = Type)) +
  geom_bar(position = "stack", stat = "identity") +
  labs(title = "Dry",
       y = "% of the total diversity",
       x = "") +
  scale_fill_manual(values = c("alpha.1" = "black",
                               "beta.1" = "darkgray",
                               "beta.2" = "gray",
                               "beta.3" = "lightgray",
                               "beta.4" = gray(0.9)),
                    labels = c("alpha.1" = "Habitat ( 1)",
                               "beta.1"  = "Habitat ( 1)",
                               "beta.2"  = "Site ( 2)",
                               "beta.3"  = "Stream ( 3)",
                               "beta.4"  = " 4")) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5), #hjust 0=left, 0.5=center, 1=right
    axis.text.x = element_text(angle = 0, hjust = 0.5),
    panel.grid = element_blank(),
  )
ggplot_bb.seco
ggsave(ggplot_bb.seco, dpi = 300, filename = "res_ms/fig-bb.seco.png")

# bb.chuva (B) #bb.chuva(B) (m_cont_bb.chuva)
bb.gchuva <- readRDS("res_ms/part-bb.gchuva.rds")
ggplot_bb.chuva <- ggplot(bb.gchuva, aes(fill = Component, y = Percentage, x = Type)) +
  geom_bar(position = "stack", stat = "identity") +
  labs(title = "Wet",
       y = "Barro Branco stream",
       x = "") +
  scale_y_continuous(position = "right") +
  scale_fill_manual(values = c("alpha.1" = "black",
                               "beta.1" = "darkgray",
                               "beta.2" = "gray",
                               "beta.3" = "lightgray",
```

```
                                "beta.4" = gray(0.9)),
                     labels = c("alpha.1" = "Habitat ( 1)",
                                "beta.1"  = "Habitat ( 1)",
                                "beta.2"  = "Site ( 2)",
                                "beta.3"  = "Stream ( 3)",
                                "beta.4"  = " 4")) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5), #hjust 0=left, 0.5=center, 1=right
    axis.text.x = element_text(angle = 0, hjust = 0.5),
    axis.text.y = element_blank(),
    panel.grid = element_blank(),
  )
ggplot_bb.chuva
ggsave(ggplot_bb.chuva, dpi = 300, filename = "res_ms/fig-bb.chuva.png")

# ca.seco (C) #ca.seco(C) (m_cont_ca.seco)
ca.gseco <- readRDS("res_ms/part-ca.gseco.rds")
ggplot_ca.seco <- ggplot(ca.gseco, aes(fill = Component, y = Percentage, x = Type)) +
  geom_bar(position = "stack", stat = "identity") +
  labs(title = "",
       y = "% of the total diversity", #\n quebra de linha
       x = "") +
  scale_fill_manual(values = c("alpha.1" = "black",
                               "beta.1" = "darkgray",
                               "beta.2" = "gray",
                               "beta.3" = "lightgray",
                               "beta.4" = gray(0.9)),
                    labels = c("alpha.1" = "Habitat ( 1)",
                               "beta.1"  = "Habitat ( 1)",
                               "beta.2"  = "Site ( 2)",
                               "beta.3"  = "Stream ( 3)",
                               "beta.4"  = " 4")) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 0, hjust = 0.5),
    panel.grid = element_blank(),
  )
ggplot_ca.seco
ggsave(ggplot_ca.seco, dpi = 300, filename = "res_ms/fig-ca.seco.png")

# ca.chuva (D) #ca.chuva(D) (m_cont_ca.chuva)
ca.gchuva <- readRDS("res_ms/part-ca.gchuva.rds")
```

```r
ggplot_ca.chuva <- ggplot(ca.gchuva, aes(fill = Component, y = Percentage, x = Type)) +
  geom_bar(position = "stack", stat = "identity") +
  labs(title = "",
       y = "Caiana stream",
       x = "") +
  scale_y_continuous(position = "right") +
  scale_fill_manual(values = c("alpha.1" = "black",
                               "beta.1" = "darkgray",
                               "beta.2" = "gray",
                               "beta.3" = "lightgray",
                               "beta.4" = gray(0.9)),
                    labels = c("alpha.1" = "Habitat ( 1)",
                               "beta.1"  = "Habitat ( 1)",
                               "beta.2"  = "Site ( 2)",
                               "beta.3"  = "Stream ( 3)",
                               "beta.4"  = " 4")) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 0, hjust = 0.5),
    axis.text.y = element_blank(),
    panel.grid = element_blank(),
  )
ggplot_ca.chuva
ggsave(ggplot_ca.chuva, dpi = 300, filename = "res_ms/fig-ca.chuva.png")

# PAINEL DE GRÁFICOS CHUVOSO E SECO

library(gridExtra)

# Organizar os gráficos em 2 linhas e 2 colunas
painel_c_s <- grid.arrange(
  ggplot_pt.seco, ggplot_pt.chuva,
  ncol = 1, nrow = 2
)

# Salvar o painel em alta resolução (300 DPI)
ggsave(
  filename = "res_ms/fig-painel_pt.cs.png",
  plot = painel_c_s,
  width = 4,
  height = 8,
  dpi = 300
)
```

```r

# PAINEL DE GRÁFICOS POR RIO

library(patchwork)

shared_scale <- scale_fill_manual(
  values = c("alpha.1" = "black",
             "beta.1"  = "darkgray",
             "beta.2"  = "gray",
             "beta.3"  = "lightgray",
             "beta.4"  = gray(0.9)),
  labels = c("alpha.1" = "Habitat ( 1)",
             "beta.1"  = "Habitat ( 1)",
             "beta.2"  = "Site ( 2)",
             "beta.3"  = "Stream ( 3)",
             "beta.4"  = " 4"),
  drop = TRUE)

p1 <- ggplot_bb.seco  + shared_scale
p2 <- ggplot_bb.chuva + shared_scale
p3 <- ggplot_ca.seco  + shared_scale
p4 <- ggplot_ca.chuva + shared_scale

painel_rios <- (p1 + p2) / (p3 + p4) +
  plot_layout(guides = "collect") &
  theme(legend.position = "right",
        legend.title = element_blank())

painel_rios

ggsave(
  filename = "res_ms/fig-painel_pt.rios.png",
  plot = painel_rios,
  width = 7,
  height = 8,
  dpi = 300
)

#TABELA DE PARTIÇÃO----

#CRIANDO UMA FUNÇÃO DE EXTRAÇÃO

extrair_linha_particao <- function(part_obj, sistema, periodo) {
```

```r

  mapa_componentes <- c(
    "alpha.1" = " 1 (Habitat)",
    "alpha.2" = " 2 (Site)",
    "alpha.3" = " 3 (Stream)",
    "beta.1"  = " 1 (Between habitats)",
    "beta.2"  = " 2 (Between sites)",
    "beta.3"  = " 3 (Between streams)",
    "gamma"   = "  (Total richness)"
  )

  stat <- part_obj$statistic
  comp_names <- names(stat)

  df <- data.frame(
    Sistema    = sistema,
    Periodo    = periodo,
    Componente = ifelse(comp_names %in% names(mapa_componentes),
                        mapa_componentes[comp_names], comp_names),
    Observado  = sprintf("%.2f%%", stat * 100),
    Esperado   = sprintf("%.2f%%", part_obj$oecosimu$means * 100),
    SES        = sprintf("%.2f", part_obj$oecosimu$z),
    p_val      = part_obj$oecosimu$pval,
    stringsAsFactors = FALSE
  )

  df$p_valor <- ifelse(df$p_val <= 0.001, "< 0.001*",
                       ifelse(df$p_val <= 0.05, sprintf("%.3f*", df$p_val),
                              sprintf("%.3f", df$p_val)))

  return(df %>% select(-p_val))
}

#PROCESSAMENTO E RUNDOWN DOS OBJETOS ADIPART

# Rodando o adipart para cada cenário e gerando os objetos corretos

#pt.total (m_cont)
#pt.seco (m_cont_s), pt.chuva (m_cont_c)
#bb.seco(A) (m_cont_bb.seco), bb.chuva(B) (m_cont_bb.chuva)
#ca.seco(C) (m_cont_ca.seco), ca.chuva(D) (m_cont_ca.chuva)

part_p_pt.total <- adipart(y = m_cont, x = pt.total, index = "richness", relative = TRUE,
```

```r
                            weights = "unif", nsimul = 999)

set.seed(666)
part_p_pt.seco <- adipart(y = m_cont_s,  x = pt.seco, index = "richness", relative = TRUE,
                          weights = "unif", nsimul = 999)
part_p_pt.chuva <- adipart(y = m_cont_c,  x = pt.chuva, index = "richness", relative = TRUE,
                          weights = "unif", nsimul = 999)

part_p_bb.seco <- adipart(y = m_cont_bb.seco,  x = bb.seco, index = "richness", relative = TR
                          weights = "unif", nsimul = 999)
part_p_bb.chuva <- adipart(y = m_cont_bb.chuva,  x = bb.chuva, index = "richness", relative =
                          weights = "unif", nsimul = 999)

part_p_ca.seco <- adipart(y = m_cont_ca.seco,  x = ca.seco, index = "richness", relative = TR
                          weights = "unif", nsimul = 999)
part_p_ca.chuva <- adipart(y = m_cont_ca.chuva,  x = ca.chuva, index = "richness", relative =
                          weights = "unif", nsimul = 999)

part_p_pt.seco <- readRDS("res_ms/part_p_pt.seco.rds")


part_p_pt.total <- readRDS("res_ms/part.p-pt.total.rds")
part_p_pt.seco <- readRDS("res_ms/part.p-pt.seco.rds")
part_p_pt.chuva <- readRDS("res_ms/part.p-pt.chuva.rds")
part_p_bb.seco <- readRDS("res_ms/part.p-bb.seco.rds")
part_p_bb.chuva <- readRDS("res_ms/part.p-bb.chuva.rds")
part_p_ca.seco <- readRDS("res_ms/part.p-ca.seco.rds")
part_p_ca.chuva <- readRDS("res_ms/part.p-ca.chuva.rds")

# CONSOLIDAÇÃO DA TABELA

tabela_raw <- rbind(
  extrair_linha_particao(part_p_pt.seco, "Partição Total (PT)", "Seco"),
  extrair_linha_particao(part_p_pt.chuva, "Partição Total (PT)", "Chuvoso"),
  extrair_linha_particao(part_p_bb.seco, "Riacho Barro Branco (BB)", "Seco"),
  extrair_linha_particao(part_p_bb.chuva, "Riacho Barro Branco (BB)", "Chuvoso"),
  extrair_linha_particao(part_p_ca.seco, "Riacho Caiana (CA)", "Seco"),
  extrair_linha_particao(part_p_ca.chuva, "Riacho Caiana (CA)", "Chuvoso")
)

# IMPRESSÃO DA TABELA GT

tabela_final <- tabela_raw %>%
```

```r
  gt(groupname_col = "Sistema") %>%
  cols_label(
    Periodo    = "Período",
    Componente = "Componente Hierárquico",
    Observado  = "Observado (%)",
    Esperado   = "Esperado (%)",
    SES        = "SES",
    p_valor    = "p-valor"
  ) %>%
  cols_align(
    align = "center",
    columns = c("Periodo", "Observado", "Esperado", "SES", "p_valor")
  )
tabela_final

# Formatar nomes das colunas diretamente
tabela_simples <- tabela_raw
colnames(tabela_simples) <- c("Sistema", "Período", "Componente", "Obs (%)", "Esp (%)", "SES

# Imprimir no console
print(tabela_simples, row.names = FALSE)

# Exportando dados para Excel
library(openxlsx)
#write.xlsx(tabela_simples, file = "res_ms/tabela de partição.xlsx", rowNames = FALSE)
wb <- loadWorkbook("res_ms/tabela de partição.xlsx")
writeData(wb, sheet = "Sheet 1", x = tabela_simples)
saveWorkbook(wb, "res_ms/tabela de partição.xlsx", overwrite = TRUE)

# Tabela final ajustada no Excel
library(readxl)
part_gttable <- read_excel(
  path  = "res_ms/tabela de partição.xlsx",
  sheet = "tabela_final",
  range = "A1:G28") #ajustar para o tamanho da tabela final
part_gttable
library(gt)
part_gttable <- gt(part_gttable)
part_gttable <- sub_missing(part_gttable,
                      columns = everything(), missing_text = "")
part_gttable
gtsave(part_gttable, "res_ms/gt-part_gttable_xlsx.html")
saveRDS(part_gttable, "res_ms/gt-part_gttable_xlsx.rds")
```

4 Discussion

Effective biodiversity conservation, whether through designing resilient networks of protected areas, quantifying species loss, or determining processes that maintain diversity, requires accounting for the spatial structuring of biological communities (Socolar et al. 2016). Fundamental to this framework is beta-diversity.

, which represents the portion of regional species richness driven by compositional dissimilarity among local assemblages. Anthropogenic pressures such as agricultural expansion, selective timber extraction, urban development, biological invasions, overexploitation, and climate change alter beta-diversity by shifting the balance between biotic heterogenization and homogenization. Although elevated beta-diversity is not inherently an ultimate conservation objective, evaluating its dynamics is critical for safeguarding regional diversity and informing strategic spatial planning.

Depth and width showed greater variations across sites as a result from the fact the streams are barred at some points from roads for local traffic and small dams for water storage. Stream width and depth showed greater variations reflecting the range of habitats along the study streams, from small and narrow streams to wider and deeper manmade reservoirs.

Bush et al. (2016); Socolar et al. (2016)

To be effective, diversity conservation must be multi-scaled, targeting local and regional processes. See (Wishart and Davies 2003, Cheung and Burrows 2024, Bedrij et al. 2026)

Why would we expect turnover during the rainy season? If all habitats become connected, wouldn't a local composition become a subset of other sites (nestedness)?Your reasoning makes complete sense from the classic perspective of hydraulic connectivity (the idea that homogenization and nestedness increase with water discharge). However, in stream ecology and specifically for zooplankton, the wet season can promote turnover (species replacement) through several well-documented ecological mechanisms: Differentiated Environmental Filtering Even though stream channels become physically connected during heavy rains, local physicochemical and hydraulic conditions remain highly heterogeneous along the stream bed: Rainfall generates pronounced spatial variation in turbidity, current velocity, depth, and nutrient/detrital inputs across different sites. Zoplankton organisms (such as rotifers, cladocerans, and copepods) exhibit contrasting tolerances to flow velocity and sediment transport. Sites featuring sheltered microhabitats (e.g., backwaters, pools, or marginal vegetation) harbor assemblage structures that are entirely distinct from open, high-velocity channels, resulting in species turnover rather than simple nestedness. Input of Benthic, Tychoplanktonic, and Riparian Species During the rainy season, increased discharge and surface runoff dislodge and wash organisms from alternative substrates into the water column: Benthic erosion and resuspension: Typically benthic, periphytic, or sediment-associated species are swept into the plankton (tychoplankton). Riparian and lateral runoff: Washing of the stream banks introduces species originating from adjacent temporary pools or newly submerged marginal vegetation. Because canopy cover, bank slope, and soil composition vary among sampling sites, distinct supplementary species

are introduced at each specific location, generating unique local community compositions (high turnover). Metacommunity Dynamics and Mass Effects According to metacommunity theory (Mass Effects): Hydrological connectivity indeed enhances organism dispersal among sites. However, when stream reaches feature contrasting local environmental conditions or distinct propagule sources, continuous dispersal can maintain local populations temporarily across different sites, thereby increasing overall compositional heterogeneity across the stream network. Heterogeneous Washout and Local Resilience Floods act as a physical hydraulic disturbance: Sampling sites do not experience the same intensity of washout (population scouring). Larger-bodied species or those poorly adapted to swift currents are selectively removed from exposed reaches, whereas fast-reproducing species or those with attachment mechanisms persist or rapidly colonize specific locations. This asynchronous response to disturbance shifts relative species composition among sites. Summary for your Manuscript Discussion: Rather than driving spatial homogenization (nestedness) solely through connectivity, the hydrological pulse functions as a key driver of spatial heterogeneity by flushing stream banks, resuspending benthic taxa, and establishing novel local hydraulic gradients. This explains why site-scale beta diversity (β_2) remained the dominant component ($\text{SES} > 0$) and deviated significantly from null expectations during the rainy season.

Freshwater habitats are among the most altered ecosystems on the planet, driven by anthropogenic activities such as improper effluent discharge, habitat destruction and degradation, and the introduction of exotic species [@altindag2025; @marcelino2025; @pinto2006; @silva2020]. In this context, it is essential to understand and evaluate how habitat structure influences the diversity and composition of aquatic fauna, such as zooplankton—a group recognized as bioindicators of pollution and eutrophication processes that provides critical ecosystem services in freshwater environments [@marcelino2025; @santos2025].

According to @cottenie2005 and @gomes2020, environmental and spatial factors play a central role in structuring biological communities, particularly in aquatic ecosystems. In open hydrological systems, such as streams, creeks, and large rivers, the continuous flow of water from headwaters to the mouth generates turbulence and suspended particle transport [@schäfer1985; @wetzel2000], influencing the composition and spatial distribution of biological communities. Variables such as water temperature, dissolved oxygen, pH, transparency, conductivity, depth, and channel width, along with other attributes associated with habitat complexity, are widely recognized as key drivers of zooplankton community structure [@silva2009; @sousa2008; @vieira2009; @wetzel2001].

The results obtained from the PCA revealed a clear differentiation among sampling units (SUs). The pattern displayed by the Barro Branco stream during the dry season, particularly at SUs located inside the biological reserve (P05 and P07), showed a stronger association with structural features such as leaf litter, as well as filamentous and attached algae. This pattern may be linked to the lower hydrological variability characteristic of the dry period, which favors structural stability [@allan2007; @allan2021]. Furthermore, low flow conditions allow submerged and marginal structures to facilitate food capture and provide spatial refuge for organisms, helping them persist in specific zones of the water column [@esteves2011]. Riparian vegetation

further enhances the diversity of underwater and bank structures, creating microhabitats that can be colonized by various zooplankton species [@cyr1988; @margalef1983; @medeiros2008].

Also during the dry season, several SUs from both Barro Branco and Caiana streams were associated with variables such as channel depth and width, presence of rocks, and submerged vegetation. These attributes, alongside water flow, are primary determinants in characterizing lotic environments [@allan2007; @bonecker1998; @fulone2008]. Depth and width measurements across both streams exhibited low variability, as expected for first-order streams under the classification of @strahler1952.

The association of certain SUs with pH, which ranged from acidic to slightly neutral, also warrants attention. According to @duré2021, pH is a key factor influencing zooplankton communities, as species display differential sensitivities to its fluctuations, ultimately impacting community composition and ecosystem functioning.

During the rainy season, the PCA demonstrated a structural rearrangement of the SUs, highlighting the strong impact of precipitation on habitat architecture. Sampling units in both Barro Branco and Caiana showed strong associations with mud, macrophytes, submerged vegetation, leaf litter, and organic debris, reflecting an increase in the input and transport of allochthonous material. Rainfall directly drives the introduction and downstream transport of allochthonous inputs into aquatic ecosystems [@granjeiro2024], enhancing environmental heterogeneity and creating favorable micro-environmental conditions for zooplankton occurrence.

The influence of riparian vegetation was particularly evident in Barro Branco, especially at SUs within the reserve boundaries. @beltrão2009 and @medeiros2008 emphasize that riparian vegetation elements strongly shape habitat structure, thereby supporting local aquatic communities. Riparian cover increases the diversity of submerged and bank structures, establishing microhabitats for zooplankton colonization [@cyr1988; @margalef1983; @medeiros2008]. Conversely, its removal can exacerbate soil erosion and sedimentation while reducing shade availability along the stream reach.

Submerged and marginal structures play a fundamental role in food entrapment and spatial shelter, assisting organisms in maintaining position within specific water-column strata [@esteves2011]. Thus, hydrological fluctuations directly mediate habitat complexity and resource availability.

Rotifera was the most taxonomically rich group across all sampling units, consistent with freshwater literature where rotifers regularly dominate in species richness and density [@almeida2009; @barbosa2018; @cabral2020; @larson2007; @majeed2023]. Rotifers are frequently characterized by high fecundity, parthenogenetic reproduction, and rapid population growth rates—traits that underpin their cosmopolitan distribution [@pourriot1997]. Feeding primarily on fine particles such as organic detritus, microalgae, and bacteria, rotifers function as r-strategists (opportunists) capable of thriving in fluctuating environments [@allan1976; @araújo2017; @dantas-silva2013; @fernández2024].

In this study, Lecanidae was the most species-rich family, mirroring broader patterns reported for freshwater zooplankton assemblages [@fernández2024; @joko2008]. Numerous studies highlight Lecanidae as the predominant rotifer family in both lotic and lentic systems [@aggio2015; @melo2011; @oliveira2009; @vieira2009]. According to @almeida2006, lecanids are primarily associated with benthic and periphytic microhabitats, particularly in macrophyte-rich areas, occurring in open water only as occasional migrants.

The occurrence of Bdelloidea across all SUs aligns with established literature emphasizing their high ecological adaptability and benthic habits [@esteves1998; @ricci2001]. Similarly, the widespread presence of *Lecane bulla* agrees with findings by @davidson2012, who noted its broad tolerance to variations in water quality and physical habitat structure.

Regarding Cladocera, the family Chydoridae exhibited the highest species richness, a pattern previously documented by @costa2018, @miranda2018, and @santos-wisniewski2002. Members of this family are typically tied to riparian vegetation and deeper benthic strata, feeding by scraping or filtering particulate organic matter [@fryer1968; @fryer1974; @santos-wisniewski2002; @smirnov1998]. As noted by @soares2011, their occurrence is strongly tied to macrophyte beds.

Among Copepoda, the only Calanoida species detected was *Notodiaptomus* sp., found exclusively in Caiana. Studies by @costa2018, @lima2022, and @melo2011 report similar low calanoid diversity in headwater streams, often capturing only single Diaptomidae representatives. Juvenile stages (nauplii and copepodids) were recorded in both streams, consistent with observations by @czerniawski2013, @czerniawski2010, and @lair2005 showing that immature copepods frequently dominate stream zooplankton drift.

In summary, presence-absence patterns clearly illustrate how dry and wet seasonal dynamics govern the composition and maintenance of zooplankton species through continuous species turnover across sampling units in both streams.

Diversity partitioning is a fundamental framework for understanding how biodiversity is distributed across spatial scales, enabling clear evaluation of species turnover [@magurran2021]. As posited by @crist2003, species diversity patterns vary across spatial and temporal dimensions, reflecting the underlying ecological drivers of community assembly. In this context, hydrological seasonality acts as a primary temporal driver shaping aquatic communities in lotic ecosystems.

Diversity partitioning is directly tied to β-diversity, which quantifies compositional dissimilarity among communities and allows extrapolation of regional diversity from localized samples [@lande1996; @oksanen2020]. β-diversity can reflect two primary phenomena: nestedness and species replacement (turnover) [@baselga2010; @freiry2019]. Nestedness occurs when species composition at lower-richness sites forms a strict subset of richer sites, reflecting non-random species loss. Conversely, turnover represents the replacement of species by different taxa along spatial and temporal gradients [@baselga2010; @baselga2012; @fleishman2017]. Communities can be structured by both processes simultaneously, with their relative contributions varying across taxonomic groups [@baselga2012]. However, these two specific sub-components were not directly quantified using specific partitioning metrics in the present study.

Spatial and temporal drivers structure species diversity across local and regional scales [@lopes2008]. Venn diagrams enabled an indirect evaluation of seasonal influences on species distributions across the Barro Branco and Caiana streams. During the dry season, overall species richness was higher ($\gamma = 58$), accompanied by a high number of unique species, particularly in Caiana. In contrast, the rainy season exhibited a decline in γ-diversity and unique species counts, whereas the number of shared species remained constant between periods.

The observed reduction in α-diversity and γ-diversity during the wet season can be attributed to seasonal hydrological pulses, including elevated discharge, increased hydraulic instability, and organismal flushing along the stream channels.

According to @lande1996, additive diversity partitioning provides a natural measure of similarity across multiple communities—specifically, the proportion of total diversity contained within local assemblages. Across the broader spatial scale (considering both streams combined), observed local alpha diversity (α) was consistently lower than expected under null models during both dry and wet seasons. Although Caiana maintained higher local species richness than Barro Branco, additive partitioning demonstrated that its local contribution was smaller than expected by chance. Low α values indicate that individual sites contribute less to regional diversity than predicted by random assembly. As noted by @heino2011, local freshwater diversity is frequently constrained by regional factors, such as dispersal limitation and habitat connectivity, which dampens the relative weight of the α component.

Habitat-scale beta diversity (β_1) was also lower than expected in both seasons, indicating that habitats within sites share a substantial portion of their zooplankton taxa. This pattern may stem from structural habitat homogeneity or high localized dispersal capabilities [@grönroos2013].

In contrast, site-scale beta diversity (β_2) exhibited observed values equal to or higher than expected, particularly during the wet season. This finding demonstrates that site-to-site variation significantly fuels total regional diversity at the local reach scale. @melo2009 and @fleishman2017 note that reach-scale β-diversity is driven by environmental heterogeneity, microhabitat availability, and localized physicochemical water variation.

Between-stream beta diversity (β_3) showed observed values significantly above null expectations in both dry and wet periods. This pattern indicates that regional processes—such as dispersal limitation, environmental filtering across catchments, and disturbance regimes—play a major role in structuring stream communities [@poorter2023].

When streams were analyzed individually, both Barro Branco and Caiana displayed low observed α-diversity. However, site-scale diversity (β_2) in Barro Branco exceeded null expectations. This pattern is likely driven by environmental heterogeneity along the stream course, dispersal dynamics, spatial scale, and species-specific responses to environmental conditions—factors well-established as drivers of lotic β-diversity [@heino2015; @heino2013]. Caiana exhibited high β_2 contributions across both seasons; additionally, β_3 during the wet season was notably high, suggesting that regional γ-diversity in this system is sustained primarily by spatial turnover.

Thus, even in streams with high local richness, total diversity remains tightly linked to spatial processes and local environmental variation [@súarez2009].

Additive partitioning effectively disentangled the differential impacts of dry and wet seasons on zooplankton assemblages. The rainy season amplifies spatial environmental heterogeneity and promotes differentiation among sites and streams as rising water levels alter flow velocities, shift physicochemical parameters, and facilitate downstream organism transport [@boyero2009; @heino2011].

In tropical streams, hydrological variability is a recognized master variable shaping aquatic communities through its control over resource availability and ecological assembly processes [@boyero2009; @golec-fialek2021; @picapedra2021].

Multiplicative partitioning is less frequently applied due to its greater interpretive complexity, making additive partitioning the preferred approach for assessing local vs. regional community drivers. Nevertheless, multiplicative partitioning offers valuable complementary insights into how α and β components scale up to regional diversity across spatial hierarchies [@jost2007; @tuomisto2010].

According to @shurin2000, zooplankton diversity is jointly governed by local and regional processes, with dispersal limitation becoming increasingly pronounced at broader geographic scales. Multiplicative partitioning results confirmed that observed α-diversity was consistently lower than expected across all spatial scales and seasons. Conversely, all β-diversity components (β_1, β_2, and β_3) were higher than expected under null expectations in both dry and wet periods. This confirms significant species composition turnover across habitats (β_1), sites (β_2), and streams (β_3).

Because local diversity (α) and beta diversity (β) can vary independently, a high α value does not automatically dictate a low or high β value, and vice versa. Consequently, high beta diversity can readily occur even when individual habitats exhibit low local richness (α) [@jost2007; @medeiros2021].

Overall, these findings demonstrate that zooplankton diversity in the Barro Branco and Caiana streams is strongly shaped by spatial processes acting at intermediate and regional scales, as evidenced by elevated β-diversity across levels. These spatial patterns highlight environmental heterogeneity and seasonal hydrological pulses as essential drivers for maintaining regional stream biodiversity.

Conclusions

Acknowledgements

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Authorship contribution statement

Authorship of this paper is based on CRediT (2026).

Ellen Gomes da Silva: Data curation, Formal analysis, Investigation, Software, Visualization, Writing – original draft.

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Ludmilla Cavalcanti Antunes Lucena: Supervision, Validation.

Thais Xavier de Melo: Supervision, Validation.

Elvio Sergio Figueredo Medeiros: Conceptualization, Data Curation, Formal Analysis, Funding Acquisition, Investigation, Methodology, Project Administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing.

Ethical approval

Activities inside the REBIO Guaribas were authorized by the Instituto Chico Mendes de Conservação da Biodiversidade (license no. SISBIO-89415-1, SISBIO-89429-2 and SISBIO-89415-2).

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this manuscript, the authors used ChatGPT (OpenAI) to assist with English language editing and improve the clarity and fluency of the writing. No AI was used for data analysis, interpretation, or generation of scientific conclusions. The authors carefully reviewed and revised all AI-generated suggestions and take full responsibility for the accuracy and content of the final manuscript (CNPq 2026).

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Figures and Tables

Figure 1: Location of sampling sites along the Barro Branco Stream (Mamanguape, PB) (Site 1 06°43'06"S and 35°10'54"W; Site 2 06°42'38"S and 35°10'38"W; Site 3 06°43'06"S and 35°10'54"W; Site 4 06°41'46"S and 35°10'36"W; Site 5 06°40'56"S and 35°10'27"; Site 6 06°40'18"S and 35°10'34"W).

Figure 2: Principal Component Analysis (PCA) biplot of the sampling units (SUs) and environmental variables in the Barro Branco and Caiana streams during the study period, categorized by stream. Points represent the SUs, whereas vectors represent the environmental variables associated with habitat structure. Ellipses represent 95% confidence regions for the distribution of the groups of points, calculated based on the t-distribution. Only variables that contributed at least 20% to the axes are shown.

Figure 3: Principal Component Analysis (PCA) biplot of the sampling units (SUs) and environmental variables in the Barro Branco and Caiana streams during the study period, categorized by season. Points represent the SUs, whereas vectors represent the environmental variables associated with habitat structure. Ellipses represent 95% confidence regions for the distribution of the groups of points, calculated based on the t-distribution. Only variables that contributed at least 20% to the axes are shown.

Figure 4: Individual-based extrapolation and rarefaction curves for the richness of resting stages of zooplanktonic taxa in each study stream. Shaded areas indicate 95% confidence intervals.

Figure 5: Classification analysis (UPGMA using Bray-Curtis dissimilarity) of wetland macroinvertebrate samples obtained in a. permanent and b. intermitent wetlands. The first number of the sample label indicates the wetland. The second number indicates the habitat type (1=hydrophytes; 2=reed-like emergent plants; and 3=leafy emergent plants). The third number indicates the individual sweeps realized in each habitat (1–3)UPGMA Hierarchical flexible beta clustering ($= -0.25$) based on Bray-Curtis distance of (a) macroinvertebrates and (b) Chironomidae in the Paraíba River during the 2015 wet season. Tag codes indicate different sections of the river: U = upper segment, M = middle segment, and T = Taperoá River segment. R = reaches (1–3), S = sites (1–3), and Hab = habitat types (1–3). Percent chaining <4%.

Figure 6: Observed and expected diversity, partitioned into one alpha and three beta components, expressed as percent of total species richness. Numbers indicate the proportion of randomized samples containing more species than the observed sample for each partition. 1: subsample species richness, 1: beta diversity among sub-samples; 2: beta diversity among pools or habitats within pool, and 3: beta diversity among river reaches.

Figure 7: Observed and expected diversity, partitioned into one alpha and three beta components, expressed as percent of total species richness. Numbers indicate the proportion of randomized samples containing more species than the observed sample for each partition. 1:

subsample species richness, 1: beta diversity among sub-samples; 2: beta diversity among pools or habitats within pool, and 3: beta diversity among river reaches.

Table 1: Environmental variables averaged for sampling sites (minimum-maximum) for the Barro Branco stream (Mamanguape, PB).

Table 2: Nutrient concentration (g L ^{1}) averaged for sampling sites (minimum-maximum) for the Barro Branco stream (Mamanguape, PB).

Table 3: List of species and counts of zooplankton collected during the 2024 hydrological cycle.

Table 4: Additive partitioning of zooplankton species richness (γ-diversity) into hierarchical alpha (α) and beta (β) diversity components across sampling levels (expressed in percentage values). Shown are the observed diversity values, mean diversity values expected under the null model, standardized effect sizes (SES), and statistical significance (P-value). Numbers in 'P-value' column indicate the proportion of randomized samples containing more species than the observed sample for each partition. α_1 (Habitat): Average species richness per habitat. β_1 (Habitat): Diversity turnover among habitats within a site. β_2 (Site): Diversity turnover among sampling sites within a stream. β_3 (Stream): Diversity turnover between streams.

Figures

Figure 1

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Tables

Table 1

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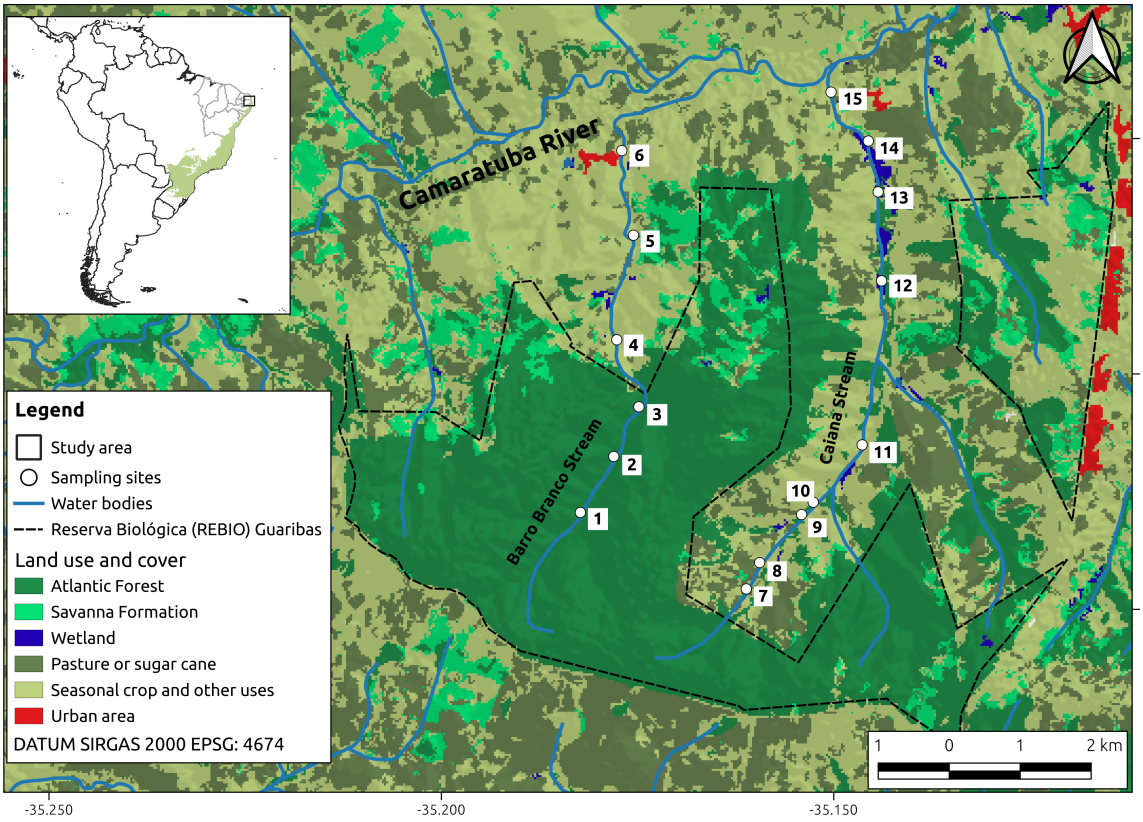


Figure 1: Study area

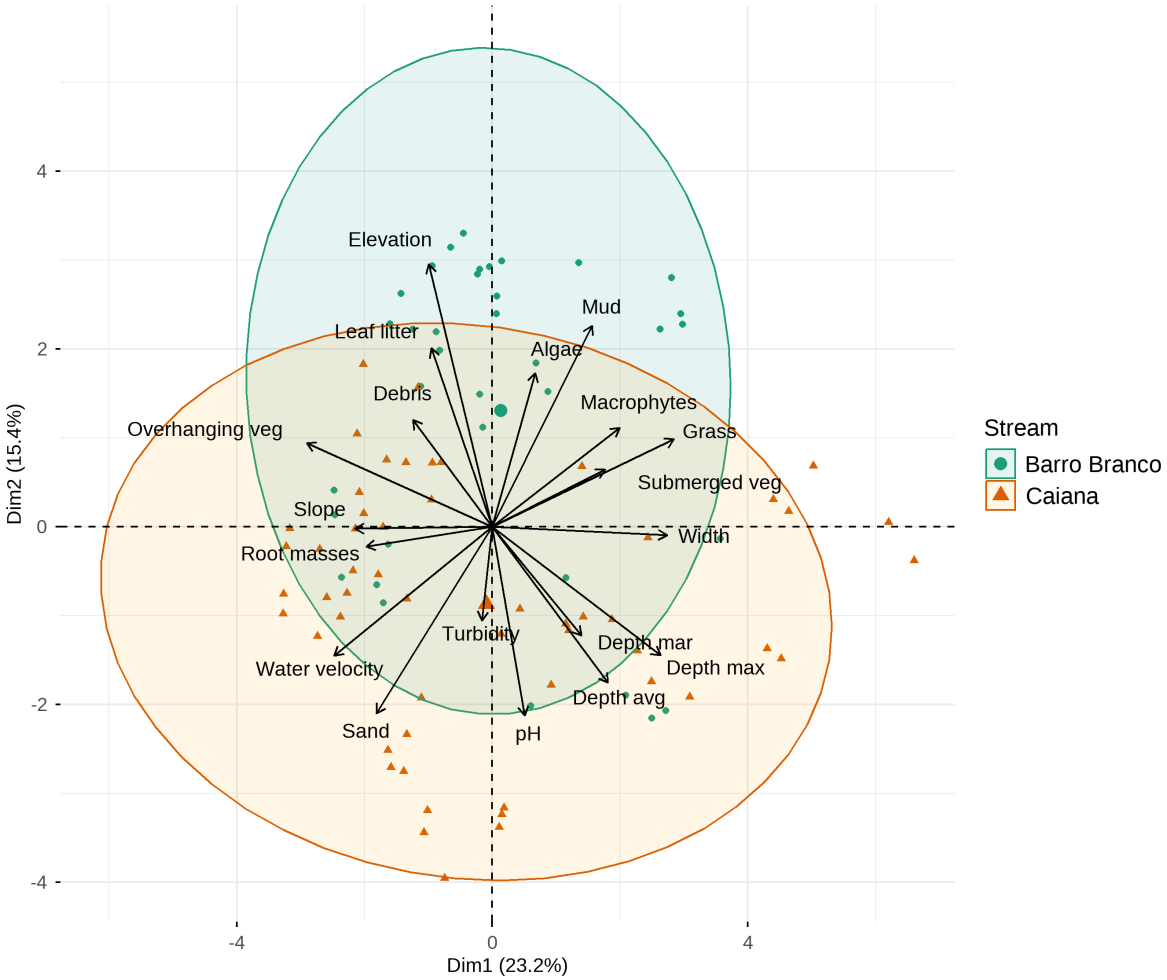


Figure 2: Principal Component Analysis, streams

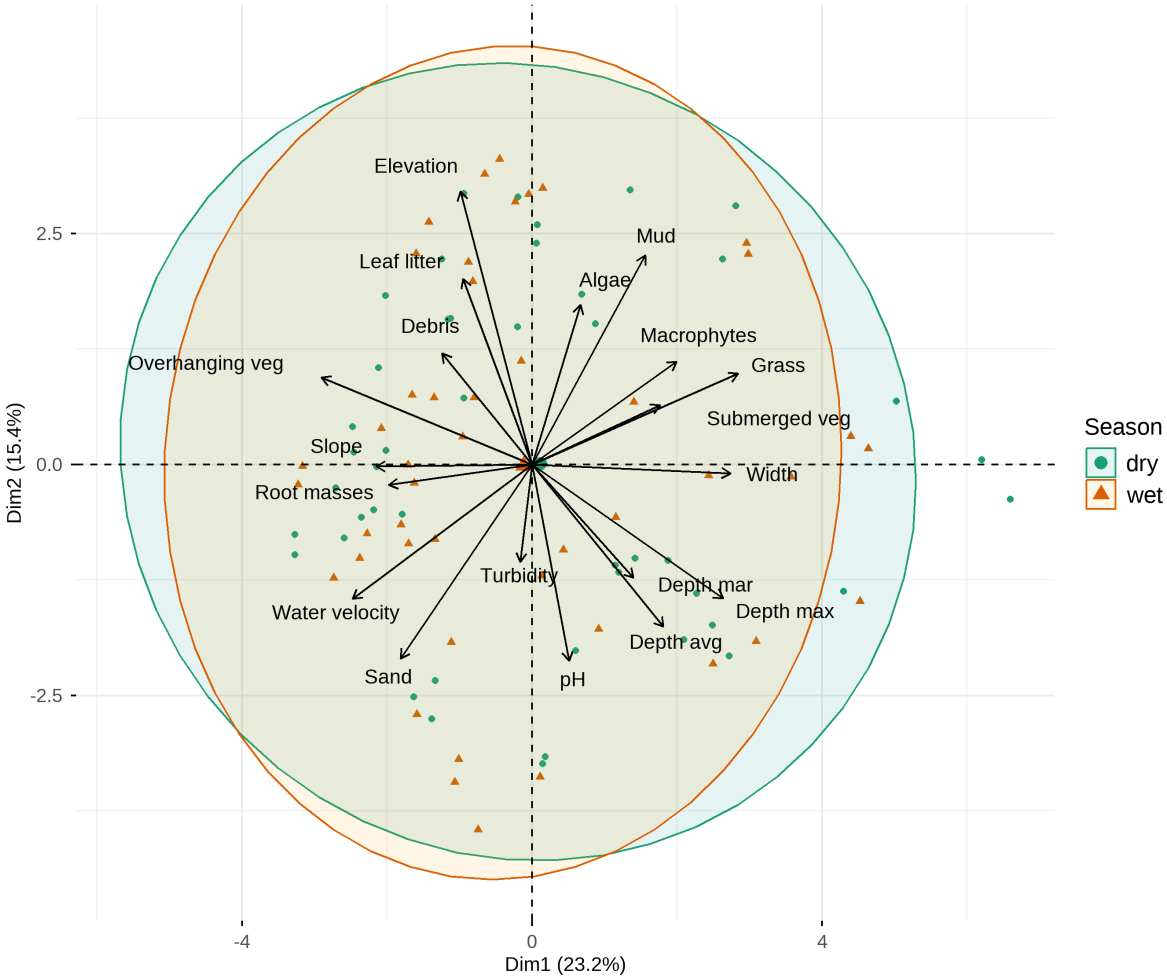


Figure 3: Principal Component Analysis, seasons

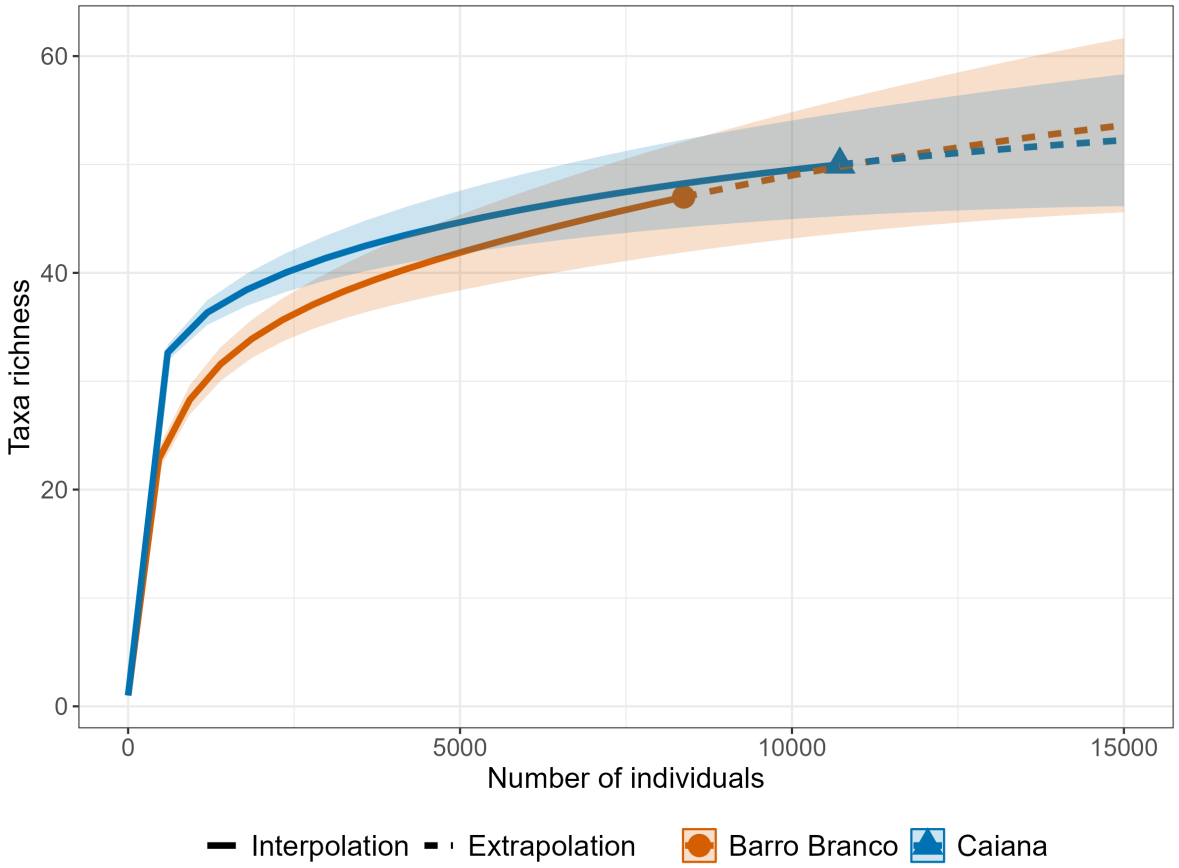


Figure 4: Accumulation curves

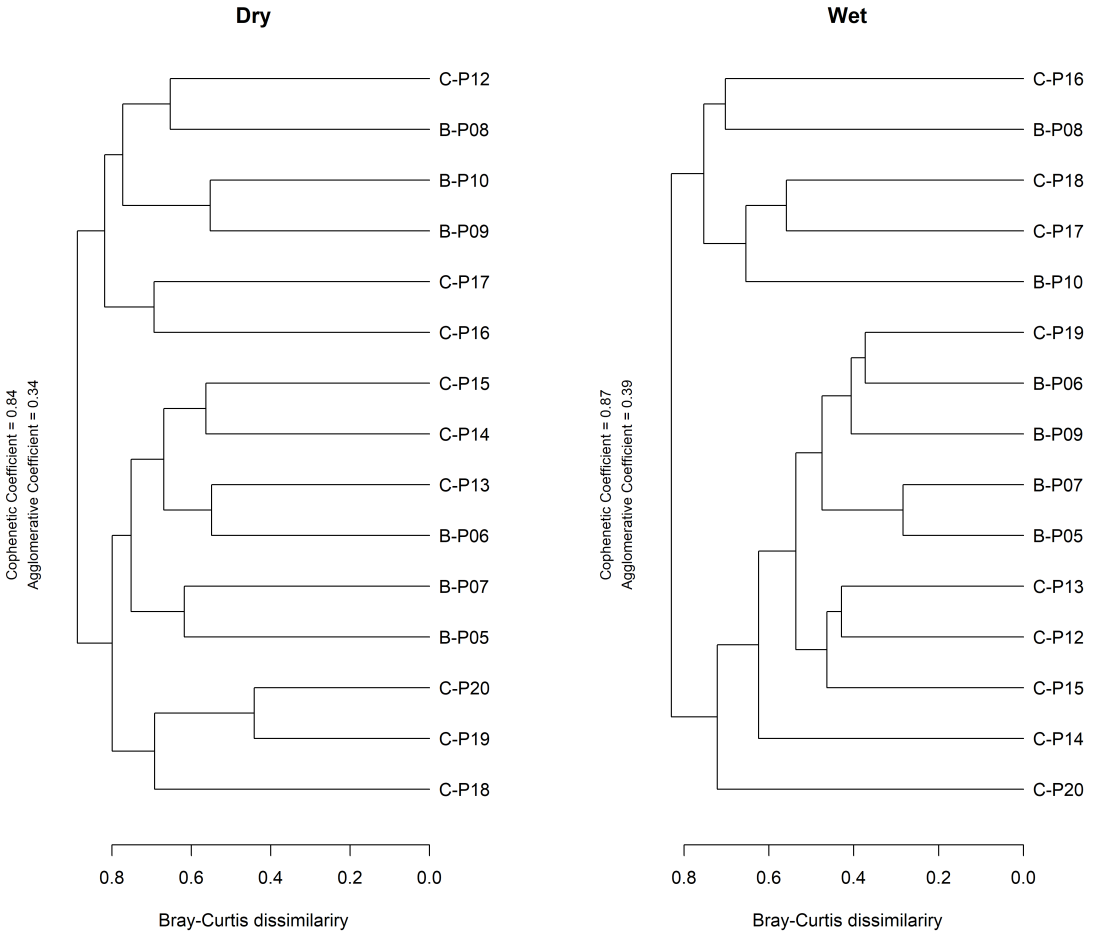


Figure 5: UPGMA Classification

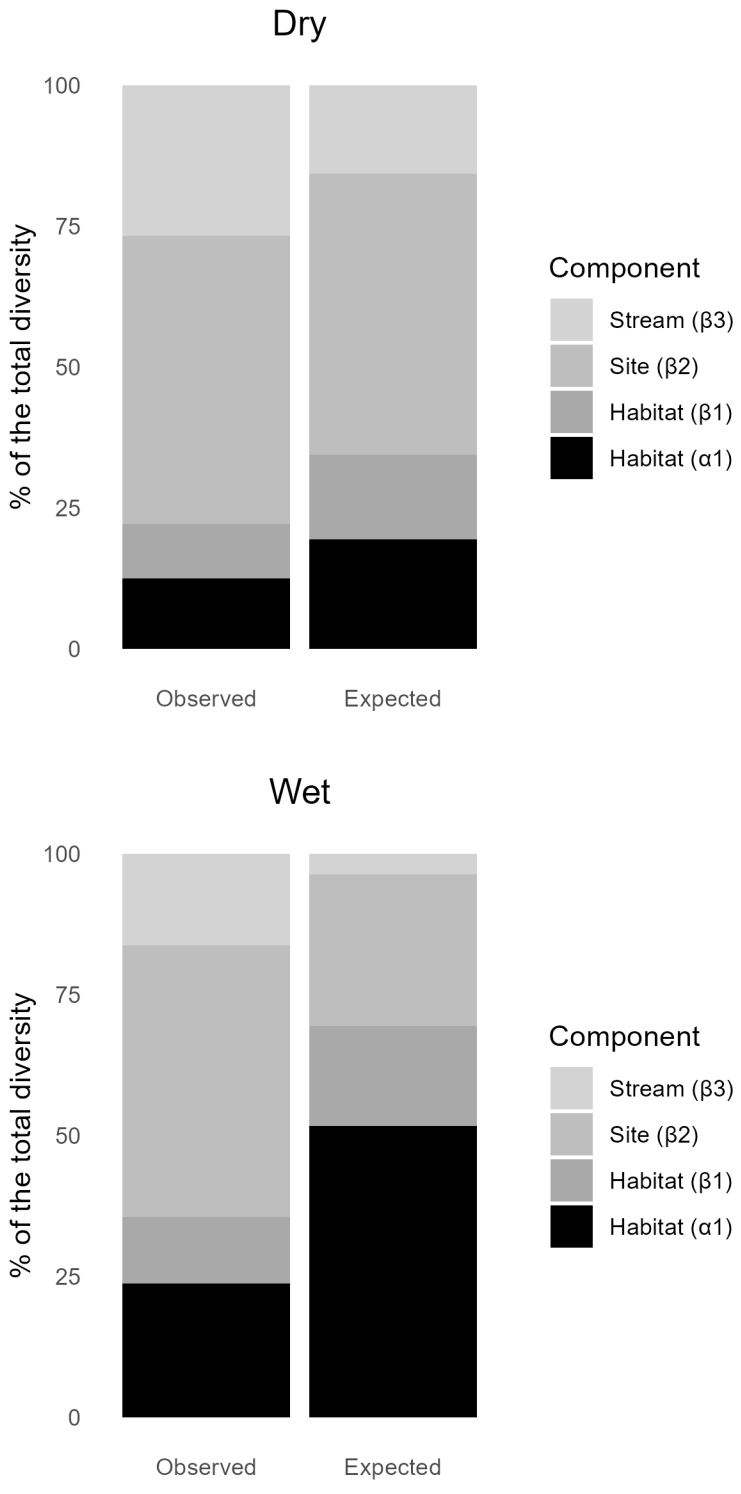


Figure 6: Partitioning plots season

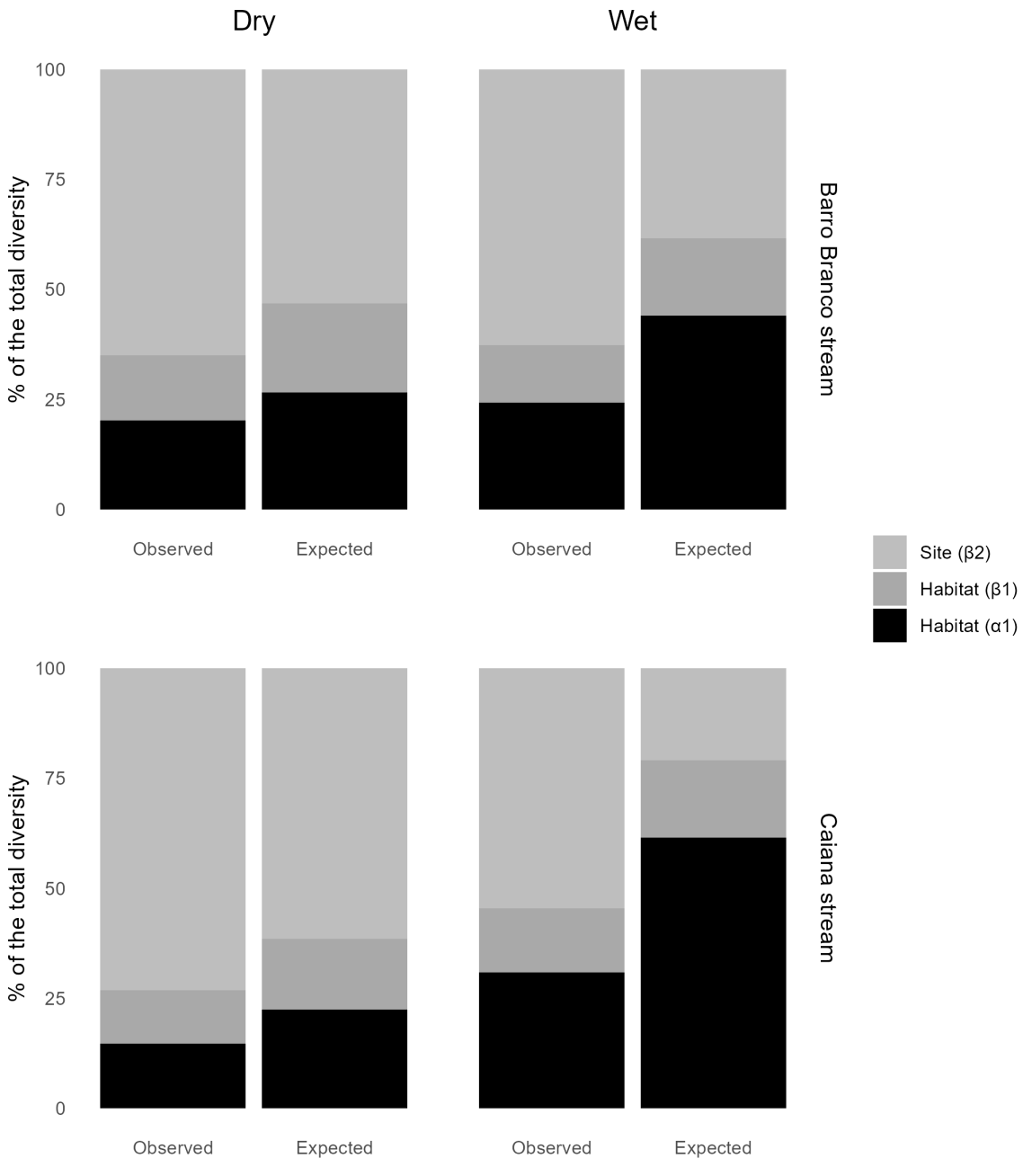


Figure 7: Partitioning plots stream

Table 1: Environment data GT table.

...1	Barro Branco stream	...3	...4	...5
	Site 1	Site 2	Site 3	Site 4
Physicochemical variables				
Temperature (°C)*	25.8 (24.9-27.3)	26.5 (25.3-27.5)	25.2 (24.9-25.5)	27.8 (27-
Dissolved oxygen (mg/L)*	8.1 (7.8-8.5)	7.5 (7.2-7.8)	7.9 (7.7-8)	7.5 (6.5-
Water velocity (m/s)	0.073 (0.025-0.161)	0.086 (0.035-0.24)	0.135 (0.022-0.207)	0.036 (0-
Conductivity (µS/cm)	137.5 (119.2-164.4)	148 (134.1-156.8)	137.5 (98.3-160.2)	154.2 (1-
Turbidity (NTU)	5 (2-10.7)	4.9 (2.1-14.1)	4.9 (1.5-9.2)	5.7 (4-6.
pH	5.9 (4.7-7.5)	5.7 (4.6-7.7)	5.1 (4.5-6)	5.5 (5-6)
Morphology				
Elevation (m a.s.l.)	107.1 (107.1-107.1)	112.8 (112.8-112.8)	89 (89-89)	89.1 (89
Slope	1.4 (1-2)	2.7 (2.5-3)	2.4 (1.5-3)	1.5 (1-2)
Average depth (cm)	12.8 (9-14)	23.7 (16.3-31.7)	23.8 (16-41)	43.3 (28
Margin depth (cm)	11.8 (3-16)	21.7 (8-30)	18.8 (7-42)	48.3 (16-
Maximum depth (cm)	15.5 (13-17)	32.7 (27-40)	29.3 (20-47)	50.8 (35-
Width (m)	6.2 (3.6-14.1)	4.3 (0.7-9.2)	2 (1.1-3.2)	7.6 (2-16
Substrate composition (%)				
Mud	100 (100-100)	100 (100-100)	95 (90-100)	74.2 (5-1
Sand	0	0	5 (0-10)	25.8 (0-9
Gravel	0	0	0	0
Rocks	0	0	0	0
Habitat structures (%)				
Aquatic macrophytes	5.8 (5-10)	10 (0-30)	0	17.9 (0-6
Algae	2.5 (0-5)	0.2 (0-1)	5.5 (0-21)	19.2 (15-
Submerged vegetation	7.5 (5-15)	8.3 (5-15)	5.8 (0-10)	5.8 (0-10
Overhanging vegetation	68.3 (40-90)	70.8 (50-85)	76.7 (60-90)	31.7 (25-
Littoral grass	3.3 (0-10)	8.3 (5-15)	6 (0-15)	27.1 (15-
Leaf litter	7.5 (5-10)	11 (1-25)	15.8 (5-30)	18.3 (10-
Root masses	5 (0-15)	10.8 (5-15)	12.5 (5-20)	4.2 (0-10
Wood debris	30.8 (20-45)	12.7 (5-20)	17.7 (5-30)	21.7 (5-3

Table 2: Nutrient concentrations at the sampling sites.

...1	Barro Branco strream	...3	...4	...5	.
	Site 1	Site 2	Site 3	Site 4	S
Ammonia (µg L ^{1})	54.8 (39.3-64.3)	58.9 (47.5-77)	72 (33.4-115.7)	81.6 (68.9-93)	8
Nitrite (µg L ^{1})	0	0	0	0	0
Nitrate (µg L ^{1})	6.0 (0-9.7)	8.8 (6.9-12.6)	5.5 (0-12.6)	18.8 (11.1-31.1)	1
Orthophosphate (µg L ^{1})	0	0	1.0 (0-3)	0	0
Phosphorus (µg L ^{1})	0	4.6 (0.3-10.3)	50.1 (0-150.3)	9 (0-27)	1

Apendices

Non-used figures and tables

Non-used codes

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1 table(1:10)
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Table 3: Checklist of taxa GT table.

...1	...2	Barro Branco stream	...4	Caian
ROTIFERA		Dry	Wet	Dry
DIGONONTA		N= 1529	N= 6840	N= 2
Bdelloidea	Bdelloidea	994	1930	835
Philodinidae				
Dissotrocha aculeata (Ehrenberg, 1832)	Dis.aculeata	0	0	2
Dicranophoridae				
Dicranophoroides sp.	Dicranop.sp	0	0	2
Notommatidae				
Cephalodella sp.	Cephalodella.sp	2	8	8
MONOGONONTA				
FLOSCULARIACEA				
Lecanidae				
Lecane aculeata (Jakubski, 1912)	Lec.aculeata	1	0	0
Lecane bulla (Gosse, 1851)	Lec.bulla	60	9	20
Lecane cornuta (Müller, 1786)	Lec.cornuta	4	4	5
Lecane crepida Harring, 1914	Lec.crepida	13	1	2
Lecane curvicornis (Murray, 1913)	Lec.curvicornis	1	5	6
Lecane furcata (Murray, 1913)	Lec.furcata	1	0	0
Lecane leontina (Turner, 1892)	Lec.leontina	3	83	7
Lecane lunaris (Ehrenberg, 1832)	Lec.lunaris	7	5	4
Lecane luna (Müller, 1776)	Lec.luna	13	4	8
Lecane monostyla (Daday, 1897):	Lec.monostyla	1	0	1
Lecane papuana (Murray, 1913)	Lec.papuana	2	0	3
Lecane quadridentata (Ehrenberg, 1830)	Lec.quadridentata	2	0	0
Lecane sp.1	Lecane.sp1	1	0	1
Lecane sp.2	Lecane.sp2	0	0	3
Lecane sp.3	Lecane.sp3	1	0	0
Brachionidae				
Brachionus patulus Müller, 1786	Bra.patulus	3	3	0
Macrochaetus sp.	Macrochaetus.sp	2	25	2
Platyias quadricornis (Ehrenberg, 1832)	Pla.quadricornis	6	176	2
Lepadellidae				
Lepadella ovalis (Müller, 1786)	Lep.ovalis	17	5	3
Lepadella patella (Müller, 1773)	Lep.patella	2	3	0
Lepadella imbricata Harring, 1914	Lep.imbricata	0	0	1
Lepadella sp.	Lepadella.sp	0	0	1
Euchlanidae				
Euchlanis dilatata Ehrenberg, 1832	Euc.dilatata	0	0	1
Synchaetidae				
Polyarthra bicerca	Pol.bicerca	0	1	36
Polyarthra dolichoptera Idelson, 1925	Pol.dolichoptera	2	0	0
Polyarthra vulgaris	Pol.vulgaris	0	1	145
Polyarthra sp.	Polyarthra.sp	8	0	0
Colurellidae				
Colurella sp.	Colurella.sp	8	0	0
Mytilinidae				
Mytilina crassipes (Lucks, 1912)	Myt.crassipes	0	0	1
Mytilina bisulcata (Lucks, 1912)	Myt.bisulcata	1	0	0

Table 4: Partition data GT table.

Partition	Season	Component	Diversity...4	Diversity...5	SES	P-valu
Partition	Season	Component	Observed (%)	Expected (%)	SES	P-valu
Season	Dry	1 (Habitat)	12.45	19.43	-22.13	< 0.00
		(Total richness)	100.00	100.00	0.00	1.000
		1 (Between habitats)	9.73	15.03	-8.93	< 0.00
		2 (Between sites)	51.09	49.81	0.64	0.519
		3 (Between streams)	26.72	15.73	5.57	< 0.00
	Wet	1 (Habitat)	23.72	51.71	-45.44	< 0.00
		(Total richness)	100.00	100.00	0.00	1.000
		1 (Between habitats)	11.78	17.68	-7.03	< 0.00
		2 (Between sites)	48.25	27.02	13.41	< 0.00
		3 (Between streams)	16.25	3.59	9.48	< 0.00
Barro Branco stream	Dry	1 (Habitat)	20.23	26.63	-11.17	< 0.00
		(Total richness)	100.00	100.00	0.00	1.000
		1 (Between habitats)	14.81	20.17	-4.53	< 0.00
		2 (Between sites)	64.96	53.19	8.94	< 0.00
	Wet	1 (Habitat)	24.24	43.98	-20.41	< 0.00
		(Total richness)	100.00	100.00	0.00	1.000
		1 (Between habitats)	13.13	17.62	-3.13	< 0.00
		2 (Between sites)	62.63	38.41	13.37	< 0.00
Caiana stream	Dry	1 (Habitat)	14.73	22.48	-15.49	< 0.00
		(Total richness)	100.00	100.00	0.00	1.000
		1 (Between habitats)	12.08	15.97	-4.18	0.003*
		2 (Between sites)	73.19	61.55	10.17	< 0.00
	Wet	1 (Habitat)	30.83	61.48	-34.59	< 0.00
		(Total richness)	100.00	100.00	0.00	1.000
		1 (Between habitats)	14.60	17.64	-2.85	0.007*
		2 (Between sites)	54.58	20.88	26.58	< 0.00